

Homework 1 – Solutions

- 1.2 (a) Since Bob has to choose exactly two options, Ω consists of the 2-element subsets of the set $\{\text{cereal}, \text{eggs}, \text{fruit}\}$:

$$\Omega = \{\{\text{cereal}, \text{eggs}\}, \{\text{cereal}, \text{fruit}\}, \{\text{eggs}, \text{fruit}\}\}$$

The items in Bob's breakfast do not come in any particular order, hence the outcomes are sets instead of ordered pairs.

- (b) The two outcomes in the event A are $\{\text{cereal}, \text{eggs}\}$ and $\{\text{cereal}, \text{fruit}\}$. In symbols,

$$A = \{\text{Bob's breakfast includes cereal}\} = \{\{\text{cereal}, \text{eggs}\}, \{\text{cereal}, \text{fruit}\}\}.$$

- 1.4 (a) This is an example of sampling with replacement, where order matters. Thus, the sample space is

$$\Omega = \{\omega = (x_1, x_2, x_3) : x_i \in \{\text{states in the U.S.}\}\}.$$

In other words, each sample point is a 3-tuple or ordered triple of U.S. states.

The problem statement contains the assumption that every day each state is equally likely to be chosen. Since $\#\Omega = 50^3 = 125,000$, each sample point ω has equal probability $P\{\omega\} = 50^{-3} = \frac{1}{125,000}$. This specifies the probability measure completely because then the probability of any event A comes from the formula $P(A) = \frac{\#A}{125,000}$.

- (b) The 3-tuple (Wisconsin, Minnesota, Florida) is a particular outcome, and hence as explained above,

$$P((\text{Wisconsin}, \text{Minnesota}, \text{Florida})) = \frac{1}{50^3}.$$

- (c) The number of ways to have Wisconsin come on Monday and Tuesday, but not Wednesday is $1 \cdot 1 \cdot 49$, with similar expressions for the other combinations. Since there is only 1 way for Wisconsin to come each of the three days, we see the total number of positive outcomes is

$$1 \cdot 1 \cdot 49 + 1 \cdot 49 \cdot 1 + 49 \cdot 1 \cdot 1 + 1 = 3 \cdot 49 + 1 = 148.$$

Thus

$$\begin{aligned} P(\text{Wisconsin's flag hung at least two of the three days}) \\ = \frac{3 \cdot 49 + 1}{50^3} = \frac{37}{31250} = 0.001184. \end{aligned}$$

- 1.6 We give two solutions, first with an ordered sample, and then without order.

- (a) Label the three green balls 1, 2, and 3, and label the yellow balls 4, 5, 6, and 7. We imagine picking the balls in order, and hence take

$$\Omega = \{(i, j) : i, j \in \{1, 2, \dots, 7\}, i \neq j\},$$

the set of ordered pairs of distinct elements from the set $\{1, 2, \dots, 7\}$. The event of two different colored balls is,

$$A = \{(i, j) : (i \in \{1, 2, 3\} \text{ and } j \in \{4, \dots, 7\}) \text{ or } (i \in \{4, \dots, 7\} \text{ and } j \in \{1, 2, 3\})\}.$$

(b) We have $\#\Omega = 7 \cdot 6 = 42$ and $\#A = 3 \cdot 4 + 4 \cdot 3 = 24$. Thus,

$$P(A) = \frac{24}{42} = \frac{4}{7}.$$

Alternatively, we could have chosen a sample space in which order does not matter. In this case the size of the sample space is $\binom{7}{2}$. There are $\binom{3}{1}$ ways to choose one of the green balls and $\binom{4}{1}$ ways to choose one yellow ball. Hence, the probability is computed as

$$P(A) = \frac{\binom{3}{1}\binom{4}{1}}{\binom{7}{2}} = \frac{4}{7}.$$

1.10 (a) We take the outcome of the experiment the number of times we roll the die (similar to the coin flip example), using ∞ to denote the outcome that we never roll a 4. Then

$$\Omega = \{\infty, 1, 2, 3, \dots\}.$$

Element k in Ω means that it took k rolls to see the first four. Element ∞ means that four never appeared.

Next we deduce the probability measure P on Ω . Since Ω is a discrete sample space (countably infinite), P is determined by giving the probabilities of all the individual sample points.

For an integer $k \geq 1$, we have

$$P(k) = P\{\text{needed } k \text{ rolls}\} = P\{\text{no fours in the first } k-1 \text{ rolls, then a 4}\}.$$

Each roll has 6 outcomes so the total number of outcomes from k rolls is 6^k . Each roll can fail to be a four in 5 ways. Hence by taking the ratio of the number of favorable outcomes over the total number of outcomes,

$$P(k) = P\{\text{no fours in the first } k-1 \text{ rolls, then a 4}\} = \frac{5^{k-1} \cdot 1}{6^k} = \left(\frac{5}{6}\right)^{k-1} \frac{1}{6}.$$

To complete the specification of the measure P , we find the value $P(\infty)$. Since the outcomes are mutually exclusive,

$$\begin{aligned} 1 = P(\Omega) &= P(\infty) + \sum_{k=1}^{\infty} P(k) \\ &= P(\infty) + \sum_{k=1}^{\infty} \left(\frac{5}{6}\right)^{k-1} \frac{1}{6} \\ &= P(\infty) + \frac{1}{6} \cdot \frac{1}{1 - 5/6} \quad (\text{geometric series}) \\ &= P(\infty) + 1. \end{aligned}$$

Thus, $P(\infty) = 0$.

(b) We already deduced above that

$$P(\text{the number four never appears}) = P(\infty) = 0.$$

Here is another solution.

$$P(\text{the number four never appears}) \leq P(\text{no fours in the first } n \text{ rolls}) = \left(\frac{5}{6}\right)^n.$$

Since $(\frac{5}{6})^n \rightarrow 0$ as $n \rightarrow \infty$ and the inequality holds for any n , the probability on the left must be zero.

- 1.11 The sample space Ω that represents the dartboard itself is a square of side length 20 inches. We can assume that the center of the board is at the origin. The event A , that the dart hits within 2 inches of the center, is then the subset of Ω described by $A = \{x : |x| \leq 2\}$. Probability is now proportional to area, and so

$$P(A) = \frac{\text{area of } A}{\text{area of the board}} = \frac{\pi \cdot 2^2}{20^2} = \frac{\pi}{100}.$$

- 1.22 (a) The sample space is the set of 52 cards. We can represent the cards with numbers from 1 to 52, or with their names. Since each outcome is equally likely, $P\{\omega\} = \frac{1}{52}$ for any fixed card ω . For any subset A of cards we have $P(A) = \frac{\#A}{52}$.
- (b) An event is a subset of the sample space Ω . In part (a) we saw that for an event A we have $P(A) = \frac{\#A}{52}$. So the desired event must have three elements. Any such set will work, for example $\{\heartsuit 2, \heartsuit 3, \heartsuit K\}$. In words, this is the event that the chosen card is the two of hearts, the three of hearts or the king of hearts.
- (c) By part (a), if $P(A) = \frac{1}{5}$ then $\frac{\#A}{52} = \frac{1}{5}$ which forces $\#A = \frac{52}{5}$. Since $\frac{52}{5}$ is not an integer, there cannot be a subset with this many elements. Consequently this probability space has no event with probability $1/5$.

- 1.28 Assume that both m and n are at least 1 so the problem is not trivial.

- (a) Sampling without replacement. We can compute the answer using either an ordered or an unordered sample. It helps to assume that the balls are labeled (e.g. by numbering them from 1 to $m+n$), although the actual labeling will not play a role in the computation. With an ordered sample we have $(m+n)(m+n-1)$ outcomes (we have $m+n$ choices for the first pick and $m+n-1$ choices for the second). The favorable outcomes can be counted by considering green-green and yellow-yellow pairs separately: their number is $m(m-1) + n(n-1)$. The answer is the ratio of the number of favorable outcomes to the total number of outcomes,

$$P\{(g,g) \text{ or } (y,y)\} = \frac{m(m-1) + n(n-1)}{(m+n)(m+n-1)}.$$

The unordered sample calculation gives the same answer:

$$P\{\text{a set of two greens or a set of two yellows}\} = \frac{\binom{m}{2} + \binom{n}{2}}{\binom{m+n}{2}} = \frac{m(m-1) + n(n-1)}{(m+n)(m+n-1)}.$$

Note: for integers $0 \leq k < \ell$, the convention is $\binom{k}{\ell} = 0$. This makes the answers above correct even if m or n or both are 1.

- (b) Sampling with replacement. Now the sample has to be ordered (there is a first pick and a second pick). The total number of outcomes is $(m+n)^2$, and the number of favorable outcomes (again counting the green-green and yellow-yellow pairs separately) is $m^2 + n^2$. This gives

$$P\{(g,g) \text{ or } (y,y)\} = \frac{m^2 + n^2}{(m+n)^2}.$$

- (c) We show two arguments for comparing the two probabilities.

First we simplify the inequality through a sequence of equivalences, by cancelling factors, multiplying away the denominators, and then cancelling some more.

$$\begin{aligned} & \text{answer to (a)} < \text{answer to (b)} \\ \iff & \frac{m(m-1) + n(n-1)}{(m+n)(m+n-1)} < \frac{m^2 + n^2}{(m+n)^2} \\ \iff & \frac{m(m-1) + n(n-1)}{m+n-1} < \frac{m^2 + n^2}{m+n} \\ \iff & (m(m-1) + n(n-1))(m+n) < (m^2 + n^2)(m+n-1) \\ \iff & (m^2 - m + n^2 - n)(m+n) < (m^2 + n^2)(m+n) - m^2 - n^2 \\ \iff & (-m-n)(m+n) < -m^2 - n^2 \\ \iff & (m+n)^2 > m^2 + n^2 \\ \iff & 2mn > 0. \end{aligned}$$

The last inequality is *always true* for positive m or n . Since the last inequality is equivalent to the first one, the first one is also *always true*.

Another way to compare the two probabilities is to compare the probabilities of the respective complements. This is just the probability of getting two balls of a different color. We can just subtract our probabilities in (a) and (b) from one, or compute the probability directly (as in 1.6).

Without replacement:

$$1 - \text{answer to (a)} = 1 - \frac{m(m-1) + n(n-1)}{(m+n)(m+n-1)} = \frac{2mn}{(m+n)(m+n-1)}$$

With replacement:

$$1 - \text{answer to (b)} = 1 - \frac{m^2 + n^2}{(m+n)^2} = \frac{2mn}{(m+n)^2}$$

These probabilities are easy to compare: since $(m+n)(m+n-1) < (m+n)(m+n)$, we have

$$1 - \text{answer to (a)} = \frac{2mn}{(m+n)(m+n-1)} > \frac{2mn}{(m+n)^2} = 1 - \text{answer to (b)},$$

which means

$$\text{answer to (a)} < \text{answer to (b)}$$

(as we saw with the first approach).

The conclusion we take from this is that if you want to maximize your chances of getting two of the same color, you want to sample with replacement rather than without replacement. One way to explain this is to note that once you remove a ball, you have diminished the chances of drawing another one of the same color.

Homework 2 – Solutions

1.12 The sample space and probability measure for this experiment were described in the solution to 1.10 in HW1 : $\Omega = \{1, 2, \dots\} \cup \{\infty\}$ with $P(\{k\}) = (\frac{5}{6})^{k-1} (\frac{1}{6})$ for positive integers k , and $P(\{\infty\}) = 0$.

(a) $P(\text{need at most 3 rolls}) = P(1) + P(2) + P(3) = \frac{1}{6}(1 + \frac{5}{6} + (\frac{5}{6})^2) = \frac{91}{216}$.

(b)

$$\begin{aligned} P(\text{even number of rolls}) &= \sum_{m=1}^{\infty} P(\{2m\}) = \sum_{m=1}^{\infty} (\frac{5}{6})^{2m-1} \frac{1}{6} = \frac{1}{5} \sum_{m=1}^{\infty} (\frac{25}{36})^m \\ &= \frac{1}{5} \cdot \frac{\frac{25}{36}}{1 - \frac{25}{36}} = \frac{5}{11}. \end{aligned}$$

1.14 From the inclusion-exclusion principle we get

$$P(A \cup B) = P(A) + P(B) - P(AB) = 0.4 + 0.7 - P(AB) = 1.1 - P(AB).$$

Rearranging this we get $P(AB) = 1.1 - P(A \cup B)$.

Since $P(A \cup B)$ is a probability, it is at most 1, so

$$P(AB) = 1.1 - P(A \cup B) \geq 1.1 - 1 = 0.1.$$

On the other hand, $B \subset A \cup B$ so $P(A \cup B) \geq P(B) = 0.7$ which gives

$$P(AB) = 1.1 - P(A \cup B) \leq 1.1 - 0.7 = 0.4.$$

Putting these together we get $0.1 \leq P(AB) \leq 0.4$.

1.16 If we see only heads, I win \$5. If we see 4 heads, I win \$3. If we see 3 heads, I win \$1. If we see 2 heads, I “win” -\$1. If we see 1 heads, I “win” -\$3. Finally, if we see 0 heads, then I “win” -\$5. Thus, the possible values of X are $\{-5, -3, -1, 1, 3, 5\}$. The sample space for the 5 coin flips is $\Omega = \{(x_1, \dots, x_5) : x_i \in \{H, T\}\}$ with $\#\Omega = 2^5$. Each individual outcome $(x_1, \dots, x_5)$ of five flips has probability 2^{-5} .

Let $k \in \{0, 1, \dots, 5\}$. To calculate the probability of exactly k heads we need to count how many five-flip outcomes yield exactly k heads. The answer is $\binom{5}{k}$, the number of ways of specifying which of the five flips are heads. Hence

$$\begin{aligned} &P(\text{precisely } k \text{ heads}) \\ &= \frac{\# \text{ ways to select } k \text{ slots from the 5 for the } k \text{ heads}}{2^5} = \binom{5}{k} 2^{-5}. \end{aligned}$$

Thus,

$$\begin{aligned}
P(X = -5) &= P(0 \text{ heads}) = 2^{-5} \\
P(X = -3) &= P(1 \text{ heads}) = 5 \cdot 2^{-5} \\
P(X = -1) &= P(2 \text{ heads}) = \binom{5}{2} \cdot 2^{-5} \\
P(X = 1) &= P(3 \text{ heads}) = \binom{5}{3} \cdot 2^{-5} \\
P(X = 3) &= P(4 \text{ heads}) = \binom{5}{4} \cdot 2^{-5} \\
P(X = 5) &= P(5 \text{ heads}) = \binom{5}{5} 2^{-5}.
\end{aligned}$$

1.18 The possible values of X are $\{3, 4, 5\}$ as these are the possible lengths of the words. The probability mass function is

$$\begin{aligned}
P(X = 3) &= P(\text{we chose one of the letters of ARE}) = \frac{3}{16} \\
P(X = 4) &= P(\text{we chose one of the letters of SOME or DOGS}) = \frac{8}{16} = \frac{1}{2} \\
P(X = 5) &= P(\text{we chose one of the letters of BROWN}) = \frac{5}{16}.
\end{aligned}$$

1.32 Here is one way to determine the number of ways to be dealt a full house. We take as our sample space the set of 5-element subsets of the deck of cards:

$$\Omega = \{\{x_1, \dots, x_5\} : x_i \in \{\text{deck of } 52\}, x_i \neq x_j \text{ if } i \neq j\}.$$

Note that $\#\Omega = \binom{52}{5}$ (and that we are using a model where we do not keep track of the order of the cards).

Let A be the event that we get a full house (three cards with the same face value and two cards with another common face value). We need $\#A$, i.e. we need to count the number of ways to get a full house. We do this by ‘constructing’ a full house, and keeping track of the number of choices during the process.

First, choose the face value for the 3 cards that share a face value. There are 13 options. Then select 3 of the 4 suits for this face value. There are $\binom{4}{3}$ ways to do that. We now have the three of a kind selected. Next, choose another face value for the remaining two cards from the remaining 12 face values. Then select 2 of the 4 suits for this face value. There are $\binom{4}{2}$ ways to do that. By the multiplication rule we conclude that there are

$$13 \cdot \binom{4}{3} \cdot 12 \cdot \binom{4}{2}$$

ways to be dealt a full house. Since there are a total of $\binom{52}{5}$ poker hands, the probability is

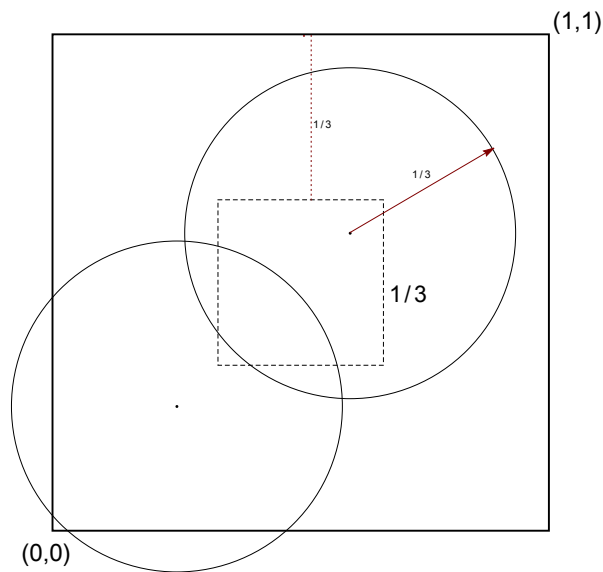
$$P(\text{full house}) = \frac{13 \cdot 12 \cdot \binom{4}{3} \binom{4}{2}}{\binom{52}{5}} \approx 0.00144.$$

1.34 Our sample space Ω is the points of the unit square $[0, 1]^2$, we measure the probability using the relative area (which is equal to the area in this case, since the unit square has area 1).

Let A be the event that the circle drawn around the chosen point is completely in the square. The circle of radius of $1/3$ around the random point is completely within the square $[0, 1]^2$ if and only if the random point lies within the smaller square with corners $(1/3, 1/3)$, $(2/3, 1/3)$, $(2/3, 2/3)$, $(1/3, 2/3)$. Hence A is exactly this smaller square of side length $1/3$.

The unit square has area one and the smaller square has area $1/9$. Consequently

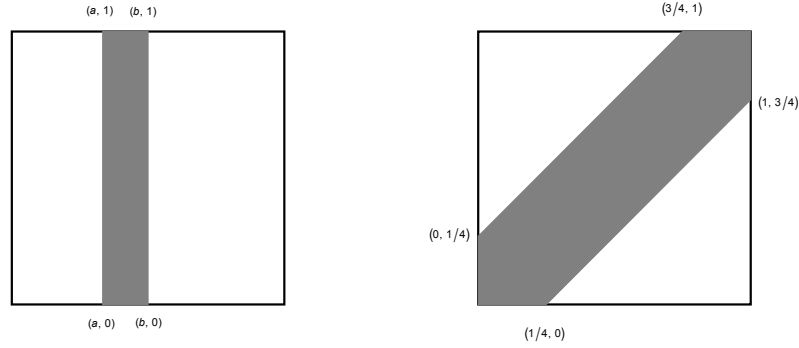
$$P(A) = P(\text{the circle lies inside the unit square}) = \frac{\text{area of } A}{\text{area of } \Omega} = \frac{1/9}{1} = \frac{1}{9}.$$



1.36 Again our sample space Ω is the points of the unit square $[0, 1]^2$, we measure the probability using the relative area (which is equal to the area in this case, since the unit square has area 1).

(a) Since (X, Y) is a uniformly random point, probability is proportional to area:

$$\begin{aligned} P(a < X < b) &= P(\text{point } (X, Y) \text{ lies in rectangle with vertices } (a, 0), (b, 0), (b, 1), (a, 1)) \\ &= \frac{\text{area of rectangle with vertices } (a, 0), (b, 0), (b, 1), (a, 1)}{\text{area of square with vertices } (0, 0), (1, 0), (1, 1), (0, 1)} \\ &= b - a. \end{aligned}$$



- (b) Let B be the event that $|X - Y| \leq 1/4$, this is the set of points (x, y) that are in the unit square for which the inequality $|x - y| \leq 1/4$ holds. This is the region between the lines $y = x - 1/4$ and $y = x + 1/4$ within the unit square (see the picture above). The area of this region B is $7/16$, this can be seen by subtracting the areas of the complementary triangles from the unit square. Thus, the desired probability $P(B)$ is also $7/16$ since the unit square has an area of one.

1.40 This is an application of inclusion-exclusion with four events. Below we use some hopefully self-evident summation notation to avoid writing out long sums.

$$\begin{aligned}
 P(\text{at least one color is repeated exactly twice}) &= P(G \cup R \cup Y \cup W) \\
 &= P(G) + P(R) + P(Y) + P(W) - \sum_{\substack{A, B \in \{G, R, Y, W\} \\ A \neq B}} P(AB) \\
 &\quad + \sum_{\substack{A, B, C \in \{G, R, Y, W\} \\ A, B, C \text{ distinct}}} P(ABC) - P(GRYW)
 \end{aligned}$$

(The first sum contains the probabilities of double intersections, the second the probabilities of triple intersections.)

Next we derive the probabilities that appear in the equation above. The sample space Ω is given by 4-tuples (x_1, x_2, x_3, x_4) where each x_i is from the set $\{\text{green, red, yellow, white}\}$. The total number of 4-tuples is $4^4 = 256$.

$$P(G) = P(\text{exactly two greens}) = \frac{\#G}{\#\Omega} = \frac{\binom{4}{2} \cdot 3 \cdot 3}{256} = \frac{27}{128}.$$

The numerator above is derived as follows: there are $\binom{4}{2}$ ways to pick the positions of the two greens in the 4-tuple. For both of the remaining two positions we have 3 colors to choose from. By the same reasoning, $P(G) = P(R) = P(Y) = P(W) = \frac{27}{128}$.

An event of type AB above means that the four draws yielded two balls of color **a** and two balls of color **b**, where **a** and **b** are two distinct particular colors. The number of 4-tuples in the event AB is $\binom{4}{2} = 6$. We can even list them easily. Here they are in lexicographic order:

aabb, abab, abba, baab, baba, bbaa.

Thus for each pair of events A, B we get $P(AB) = 6/256 = 3/128$.

Events of the type ABC are empty because four draws cannot yield three different colors that each appear exactly twice. For the same reason $GRYW = \emptyset$.

Putting everything together gives

$$\begin{aligned} &P(\text{at least one color is repeated exactly twice}) \\ &= 4 \cdot \frac{27}{128} - 6 \cdot \frac{3}{128} = \frac{45}{64} \approx 0.7031. \end{aligned}$$

Another solution: You can also get the answer by counting the favorable outcomes directly, although some care is needed. If we have at least one color repeated exactly twice then we have one or two colors that are repeated twice. (We cannot have more than two because we only have four picks.) Let A_1 be the event that there is exactly one color that is repeated twice (the four picks are of the form $\mathbf{a}, \mathbf{a}, \mathbf{b}, \mathbf{c}$ in some order), and A_2 the event that there are exactly two colors that are repeated twice (the four picks are of the form $\mathbf{a}, \mathbf{a}, \mathbf{b}, \mathbf{b}$ in some order).

We can uniquely describe an element (x_1, x_2, x_3, x_4) of A_2 by giving the color of x_1 (4 choices), the position of the other ball with the same color as x_1 (3 choices), and the other color that shows up besides x_1 's color (3 choices). This gives $\#A_2 = 4 \cdot 3 \cdot 3 = 36$.

An element (x_1, x_2, x_3, x_4) of A_1 can be uniquely described by identifying the repeated color (4 choices), the positions where this color shows up ($\binom{4}{2} = 6$ choices), and choosing the colors of the remaining two balls in order (3 and 2 choices, respectively). This gives $\#A_1 = 4 \cdot \binom{4}{2} \cdot 3 \cdot 2 = 144$. Hence

$$\#\text{at least one color is repeated exactly twice} = 36 + 144 = 180$$

and the corresponding probability is $\frac{180}{4^4} = \frac{45}{64}$.

Homework 3 – Solutions

2.4 Let A be the event that we picked the ball labeled 5 and B the event that we picked the first urn. Then we have $P(B) = 1/2$, $P(B^c) = P(\text{we picked the second urn}) = 1/2$. Moreover, from the setup of the problem

$$P(A|B) = P(\text{we chose the number 5} \mid \text{we chose from the first urn}) = 0,$$

$$P(A|B^c) = P(\text{we chose the number 5} \mid \text{we chose from the second urn}) = \frac{1}{3}.$$

We compute $P(A)$ by conditioning on B and B^c :

$$P(A) = P(A|B)P(B) + P(A|B^c)P(B^c) = 0 \cdot \frac{1}{2} + \frac{1}{3} \cdot \frac{1}{2} = \frac{1}{6}.$$

Another way to see this is to note that $A = AB^c$, since we can only choose 5 if we picked from the second urn. Then

$$P(AB^c) = P(A|B^c)P(B^c) = \frac{1}{3} \cdot \frac{1}{2} = \frac{1}{6}.$$

2.10 Define the following events:

$$A = \{\text{outcome of the roll is 4}\} \quad \text{and} \quad B_k = \{\text{the } k\text{-sided die is picked}\}, \quad k \in \{4, 6, 12\}.$$

We have

$$P(B_4) = P(B_6) = P(B_{12}) = \frac{1}{3},$$

since each of the three dice are equally likely to be picked.

The conditional probability of rolling a 4 given that we picked a k -sided die is $1/k$, hence

$$P(A|B_k) = \frac{1}{k}, \quad k \in \{4, 6, 12\}.$$

Hence using Bayes' formula gives

$$\begin{aligned} P(B_6|A) &= \frac{P(A \cap B_6)}{P(A)} = \frac{P(A|B_6)P(B_6)}{P(A|B_4)P(B_4) + P(A|B_6)P(B_6) + P(A|B_{12})P(B_{12})} \\ &= \frac{\frac{1}{6} \cdot \frac{1}{3}}{\frac{1}{4} \cdot \frac{1}{3} + \frac{1}{6} \cdot \frac{1}{3} + \frac{1}{12} \cdot \frac{1}{3}} = \frac{1}{3}. \end{aligned}$$

2.14 For disjoint events A, B we have $AB = \emptyset$ and $P(AB) = P(\emptyset) = 0$. Independence requires $P(A)P(B) = P(AB) = 0$, which can only happen if $P(A)$ or $P(B)$ is equal to 0. Hence disjoint events A and B are independent if and only if at least one of them has probability zero.

2.16 Let us label heads as 0 and tails as 1. The sample space is

$$\Omega = \{(s_1, s_2, s_3) : \text{each } s_i \in \{0, 1\}\},$$

the set of ordered triples of zeros and ones. $\#\Omega = 8$ and so for equally likely outcomes we have $P(\omega) = 1/8$ for each $\omega \in \Omega$. The events and their probabilities we need for answering the question of independence are

$$\begin{aligned} P(A_1) &= P\{(0,0,0), (0,0,1), (0,1,0), (0,1,1)\} = \frac{4}{8} = \frac{1}{2}, \\ P(A_2) &= P\{(0,1,0), (0,1,1), (1,0,0), (1,0,1)\} = \frac{4}{8} = \frac{1}{2}, \\ P(A_3) &= P\{(0,1,1), (1,0,1), (1,1,0), (0,0,0)\} = \frac{4}{8} = \frac{1}{2}, \\ P(A_1A_2) &= \{(0,1,0), (0,1,1)\} = \frac{2}{8} = \frac{1}{4} = \frac{1}{2} \cdot \frac{1}{2} = P(A_1)P(A_2), \\ P(A_1A_3) &= \{(0,1,1), (0,0,0)\} = \frac{2}{8} = \frac{1}{4} = \frac{1}{2} \cdot \frac{1}{2} = P(A_1)P(A_3), \\ P(A_2A_3) &= \{(0,1,1), (1,0,1)\} = \frac{2}{8} = \frac{1}{4} = \frac{1}{2} \cdot \frac{1}{2} = P(A_2)P(A_3), \\ P(A_1A_2A_3) &= \{(0,1,1)\} = \frac{1}{8} = \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = P(A_1)P(A_2)P(A_3). \end{aligned}$$

All the four possible combinations of more than two events from A_1, A_2, A_3 satisfy the product identity. Hence independence of A_1, A_2, A_3 has been verified.

2.18 There are 90 numbers to choose from and so each singleton $\{a\}$ with $11 \leq a \leq 99$ has probability $\frac{1}{90}$.

(a) From enumerating the possible values of X , we see that $P(X = k) = \frac{1}{9}$ for each $k \in \{1, 2, \dots, 9\}$. (For example, the event $\{X = 3\} = \{30, 31, \dots, 39\}$ has 10 outcomes from the 90 total.) For Y we have $P(Y = \ell) = \frac{1}{10}$ for each $\ell \in \{0, 1, 2, \dots, 9\}$. (For example, the event $\{Y = 3\} = \{13, 23, 33, \dots, 93\}$ has 9 outcomes from the 90 total.)

The intersection $\{X = k, Y = \ell\}$ contains exactly one number from the 90 outcomes, namely $10k + \ell$. (For example $\{X = 3, Y = 5\} = \{35\}$). Thus for each pair (k, ℓ) of possible values,

$$P(X = k, Y = \ell) = P\{10k + \ell\} = \frac{1}{90} = \frac{1}{9} \cdot \frac{1}{10} = P(X = k)P(Y = \ell).$$

Thus we have checked that X and Y are independent.

(b) To show that independence fails, we need to find only one case where the product property $P(X = k, Z = m) = P(X = k)P(Z = m)$ fails. Let's take an extreme case. The smallest possible value for Z is 1 that comes only from the outcome 10, since the sum of the digits is $1 + 0 = 1$. (Formally, since Z is a function on Ω , $Z(10) = 1 + 0 = 1$.) And so $P(Z = 1) = P\{10\} = \frac{1}{90}$. If we take $X = 2$, we cannot get $Z = 1$. Here is the precise derivation:

$$P(X = 2, Z = 1) = P(\{20, 21, \dots, 29\} \cap \{10\}) = P(\emptyset) = 0.$$

Since $P(X = 2)P(Z = 1) = \frac{1}{9} \cdot \frac{1}{90} = \frac{1}{810} \neq 0$, we have shown that X and Z are not independent.

2.30 Identical twins have the same gender. We assume that identical twins are equally likely to be boys or girls. We assume that fraternal twins are also equally likely to be boys or girls, but independently of each other. Thus fraternal twins are two girls with probability $\frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$. Let I be the event that the twins are identical, $F = I^c$ the event that the twins are fraternal, and let A be the event that Miranda has two girls.

Then we have $P(I) = 1/3$, $P(F) = 2/3$, $P(A|I) = 1/2$ and $P(A|F) = 1/4$.

$$(a) P(\text{two girls}) = P(A) = P(AI) + P(AF) = P(A|I)P(I) + P(A|F)P(F) = \frac{1}{2} \cdot \frac{1}{3} + \frac{1}{4} \cdot \frac{2}{3} = \frac{1}{3}.$$

We used the fact that $I, F = I^c$ form a partition.

$$(b) P(I|A) = \frac{P(A|I)P(I)}{P(A)} = \frac{\frac{1}{2} \cdot \frac{1}{3}}{\frac{1}{3}} = \frac{1}{2}. \text{ (Where we used part (a) for } P(A).)$$

2.34 Since the urns are interchangeable, we can put the marked ball in urn 1. There are three ways to arrange the two unmarked balls. Let case i for $i \in \{0, 1, 2\}$ denote the situation where we put i unmarked balls together with the marked ball, and the remaining $2 - i$ unmarked balls in the other urn. Let M denote the event that your friend draws the marked ball, and A_j the event that she chooses urn j , $j = 1, 2$. Since $P(M|A_2) = 0$, we get the following probabilities.

$$\text{Case 0: } P(M) = P(M|A_1)P(A_1) = 1 \cdot \frac{1}{2} = \frac{1}{2}.$$

$$\text{Case 1: } P(M) = P(M|A_1)P(A_1) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}.$$

$$\text{Case 2: } P(M) = P(M|A_1)P(A_1) = \frac{1}{3} \cdot \frac{1}{2} = \frac{1}{6}.$$

So for (a) you would put all the balls in one urn (Case 2) while for (b) she would put the marked ball in one urn and the other balls in the other urn.

(c) The situation is analogous. If we put k unmarked balls together with the marked ball in urn 1, then

$$P(M) = P(M|A_1)P(A_1) = \frac{1}{k+1} \cdot \frac{1}{2} = \frac{1}{2(k+1)}.$$

Hence to minimize the chances of drawing the marked ball, put all the balls in one urn, and to maximize the chances of drawing the marked ball, put the marked ball in one urn and all the unmarked balls in the other.

2.52 Yes, A, B , and C are mutually independent. There are four equations to check:

$$(i) P(AB) = P(A)P(B)$$

$$(ii) P(AC) = P(A)P(C)$$

$$(iii) P(BC) = P(B)P(C)$$

$$(iv) P(ABC) = P(A)P(B)P(C).$$

(i) comes from inclusion-exclusion:

$$P(AB) = P(A) + P(B) - P(A \cup B) = 0.06 = P(A)P(B).$$

(ii) comes from $P(AC) = P(C) - P(A^c C) = 0.03 = P(A)P(C)$. (iii) is given. Finally, (iv) comes from using inclusion-exclusion once more and the previous computations:

$$\begin{aligned} P(ABC) &= P(A \cup B \cup C) - P(A) - P(B) - P(C) \\ &\quad + P(AB) + P(AC) + P(BC) \\ &= 0.006 = P(A)P(B)P(C). \end{aligned}$$

2.54 (a) It is possible. We use the fact that $A = AB \cup AB^c$ and that these are mutually exclusive:

$$\begin{aligned} P(A) &= P(AB) + P(AB^c) = P(A|B)P(B) + P(A|B^c)P(B^c) \\ &= \frac{1}{3}P(B) + \frac{1}{3}P(B^c) = \frac{1}{3}(P(B) + P(B^c)) = \frac{1}{3}. \end{aligned}$$

(b) A and B are independent. By part (a) and the given information,

$$P(A) = P(A|B) = \frac{P(AB)}{P(B)}$$

from which $P(AB) = P(A)P(B)$ and independence has been verified. (Note that the value $\frac{1}{3}$ was not needed for this conclusion.)

Homework 4 – Solutions

- 2.20 (a) Let S_5 denote the number of threes in the first five rolls. This random variable has Binomial(5, 1/6) distribution. (It counts successes in 5 independent trials, the success is rolling a three.) Then

$$P(S_5 \leq 2) = \sum_{k=0}^2 P(S_5 = k) = \sum_{k=0}^2 \binom{5}{k} \left(\frac{1}{6}\right)^k \left(\frac{5}{6}\right)^{5-k}.$$

- (b) Let N be the number of rolls needed to see the first three. This is a Geometric(1/6) distributed random variable (we are waiting for the first success). From the p.m.f. of a geometric random variable,

$$P(N > 4) = \sum_{k=5}^{\infty} P(N = k) = \sum_{k=5}^{\infty} \left(\frac{5}{6}\right)^{k-1} \left(\frac{1}{6}\right) = \left(\frac{5}{6}\right)^4.$$

Equivalently,

$$P(N > 4) = P(\text{no threes within the first four rolls}) = \left(\frac{5}{6}\right)^4.$$

- (c) We can approach this in a couple different ways. ,

$$P(5 \leq N \leq 19) = \sum_{k=5}^{19} P(N = k) = \sum_{k=5}^{19} \left(\frac{5}{6}\right)^{k-1} \left(\frac{1}{6}\right)$$

We can evaluate this expression using the result of part (b):

$$P(5 \leq N \leq 19) = P(N > 4) - P(N > 19) = \left(\frac{5}{6}\right)^4 - \left(\frac{5}{6}\right)^{19}.$$

We can also use the independence of rolls to get the same answer directly:

$$\begin{aligned} P(5 \leq N \leq 19) &= P(\text{no three in the first four rolls, at least one three in rolls 5–19}) \\ &= \left(\frac{5}{6}\right)^4 \left(1 - \left(\frac{5}{6}\right)^{15}\right) = \left(\frac{5}{6}\right)^4 - \left(\frac{5}{6}\right)^{19}. \end{aligned}$$

Here we used

$$P(\text{at least one three in rolls 5–19}) = 1 - P(\text{no threes in rolls 5–19}) = 1 - \left(\frac{5}{6}\right)^{15}.$$

- 2.58 (a) We will show that these events are pairwise independent. We will show that A and B are independent, the argument is the same for the other pairs. We need to show that $P(AB) = P(A)P(B)$. We label the days with numbers from 1 to 365. Then our sample space is

$$\Omega = \{(a, b, c) : a, b, c \in \{1, 2, \dots, 365\}\},$$

the size of the sample space is $\#\Omega = 365^3$.

AB is the event that $a = b$ and $b = c$, which is just $a = b = c$. There are 365 such triples, so

$$\#AB = \{\text{all three have same birthday}\} = 365, \quad P(AB) = \frac{365}{365^3} = \frac{1}{365^2}$$

Also,

$$\#A = \{\text{Alex and Betty have the same birthday}\} = 365 \cdot 1 \cdot 365,$$

where we counted as follows: we have 365 choices for the birthday of Alex, then only once choice for Betty, and then another 365 choices for Conlin. Thus,

$$P(A) = \frac{365^2}{365^3} = \frac{1}{365}.$$

Similarly, $P(B) = \frac{1}{365}$ and so,

$$P(AB) = P(A)P(B).$$

(b) The events are not independent. Note that $ABC = AB$ and so,

$$P(ABC) = P(AB) = \frac{1}{365^2} \neq P(A)P(B)P(C) = \frac{1}{365^3}.$$

2.61 (a) Treat each draw as a trial: green is success, red is failure. By counting favorable outcomes, the probability of success is $p = \frac{3}{7}$ for each draw. Because we draw with replacement the outcomes are independent. Thus the number of greens in the 9 picks is the number of successes in 9 trials, hence a $\text{Bin}(9, \frac{3}{7})$ distribution. Using the probability mass function of the binomial distribution gives

$$P(X \geq 1) = 1 - P(X = 0) = 1 - (1 - p)^9 \approx 0.9935,$$

$$P(X \leq 5) = \sum_{k=0}^5 P(X = k) = \sum_{k=0}^5 \binom{9}{k} p^k (1 - p)^{9-k} \approx 0.8653.$$

(b) N is the number of trials needed for the first success, and so has geometric distribution with parameter $p = \frac{3}{7}$. The probability mass function of the geometric distribution gives

$$P(N \leq 9) = \sum_{k=1}^9 P(N = k) = \sum_{k=1}^9 p(1 - p)^{k-1} \approx 0.9935.$$

You can also compute this by considering the complement:

$$P(N \leq 9) = 1 - P(N > 9) = 1 - P(\text{first 9 trials are failures}) = 1 - (1 - p)^9$$

(c) We have $P(X \geq 1) = P(N \leq 9)$. We can check this by using the geometric sum formula (or the second solution in part (b)) to get

$$\sum_{k=1}^9 p(1 - p)^{k-1} = p \frac{1 - (1 - p)^9}{1 - (1 - p)} = 1 - (1 - p)^9.$$

Here is another way to see this, without any algebra. Imagine that we draw balls with replacement infinitely many times. Think of X as the number of green balls in the first 9

draws. N is still the number of draws needed for the first green. Now if $X \geq 1$, then we have at least one green within the first 9 draws, which means that the first green draw happened within the first 9 draws. Thus $X \geq 1$ implies $N \leq 9$. But this works in the opposite direction as well: if $N \leq 9$ then the first green draw happened within the first 9 draws, which means that we must have at least one green within the first 9 picks. Thus $N \leq 9$ implies $X \geq 1$. This gives the equality of the two events: $\{X \geq 1\} = \{N \leq 9\}$, and hence the probabilities must agree as well.

2.70 The relevant probabilities: $P(A) = P(B) = 2p(1-p)$ and

$$P(AB) = P\{(T, H, T), (H, T, H)\} = p^2(1-p) + p(1-p)^2 = p(1-p).$$

Thus A and B are independent if and only if

$$\begin{aligned} (2p(1-p))^2 &= p(1-p) \iff 4p^2(1-p)^2 - p(1-p) = 0 \\ &\iff p(1-p)(4p(1-p) - 1) = 0 \\ &\iff p = 0 \text{ or } 1-p = 0 \text{ or } 4p(1-p) - 1 = 0 \iff p \in \{0, \frac{1}{2}, 1\}. \end{aligned}$$

Note that cancelling $p(1-p)$ from the very first equation misses the solutions $p = 0$ and $p = 1$.

2.71 Let $F = \{\text{coin is fair}\}$, $B = \{\text{coin is biased}\}$ and $A_k = \{k\text{th flip is tails}\}$. We assume that conditionally on F , the events A_k are independent, and similarly conditionally on B . Let $D_n = A_1 \cap A_2 \cap \dots \cap A_n = \{\text{the first } n \text{ flips are all tails}\}$.

(a)

$$\begin{aligned} P(B|D_n) &= \frac{P(D_n|B)P(B)}{P(D_n|B)P(B) + P(D_n|F)P(F)} = \frac{(\frac{3}{5})^n \frac{1}{10}}{(\frac{3}{5})^n \frac{1}{10} + (\frac{1}{2})^n \frac{9}{10}} \\ &= \frac{(\frac{3}{5})^n}{(\frac{3}{5})^n + 9(\frac{1}{2})^n}. \end{aligned}$$

In particular, $P(B|D_1) = \frac{2}{17}$ and $P(B|D_2) = \frac{4}{29}$.

(b) You can compute the numerical values for various n to get

$$\frac{(\frac{3}{5})^{24}}{(\frac{3}{5})^{24} + 9(\frac{1}{2})^{24}} \approx 0.898$$

while

$$\frac{(\frac{3}{5})^{25}}{(\frac{3}{5})^{25} + 9(\frac{1}{2})^{25}} \approx 0.914,$$

so 25 flips are needed.

You can also get the answer by simplifying the probability in part (a):

$$\frac{(\frac{3}{5})^n}{(\frac{3}{5})^n + 9(\frac{1}{2})^n} = \frac{1}{1 + 9(\frac{5}{6})^n}$$

and then solving $\frac{1}{1+9(\frac{5}{6})^n} \geq 0.9$ to get $(5/6)^n \geq 1/81$ and $n \geq \frac{\log(81)}{\log(\frac{6}{5})} \approx 24.1027$.

(c)

$$\begin{aligned} P(A_{n+1}|D_n) &= \frac{P(D_{n+1})}{P(D_n)} = \frac{P(D_{n+1}|B)P(B) + P(D_{n+1}|F)P(F)}{P(D_n|B)P(B) + P(D_n|F)P(F)} \\ &= \frac{\left(\frac{3}{5}\right)^{n+1}\frac{1}{10} + \left(\frac{1}{2}\right)^{n+1}\frac{9}{10}}{\left(\frac{3}{5}\right)^n\frac{1}{10} + \left(\frac{1}{2}\right)^n\frac{9}{10}}. \end{aligned}$$

3.2 (a) We must have that the probability mass function sums to one. Hence, we require

$$1 = \sum_{k=1}^6 p(k) = c(1 + 2 + 3 + 4 + 5 + 6) = 21c.$$

Thus, $c = \frac{1}{21}$.

(b) The probability that X is odd is

$$P(X \text{ is odd}) = P(X \in \{1, 3, 5\}) = p(1) + p(3) + p(5) = \frac{1}{21}(1 + 3 + 5) = \frac{9}{21} = \frac{3}{7}.$$

3.4 (a) The density f of X is $\frac{1}{6}$ on $[4, 10]$ and zero otherwise. Hence,

$$P(X < 6) = \int_{-\infty}^6 f(x)dx = \int_4^6 \frac{1}{6}dx = \frac{6-4}{6} = \frac{1}{3}.$$

(b)

$$\begin{aligned} P(|X - 7| > 1) &= P(X - 7 > 1) + P(X - 7 < -1) = P(X > 8) + P(X < 6) \\ &= \int_8^{\infty} f(x)dx + \int_{-\infty}^6 f(x)dx = \int_8^{10} f(x)dx + \int_4^6 f(x)dx \\ &= \frac{10-8}{6} + \frac{6-4}{6} = \frac{2}{3}. \end{aligned}$$

(c) By definition of the conditional probability we have

$$P(X < t | X < 6) = \frac{P(X < t, X < 6)}{P(X < 6)}.$$

For $4 \leq t \leq 6$ then $\{X < t, X < 6\}$ is just $\{X < t\}$, hence we have

$$P(X < t, X < 6) = P(X < t) = \int_{-\infty}^t f(x)dx = \int_4^t \frac{1}{6}dx = \frac{t-4}{6},$$

and this leads to

$$P(X < t | X < 6) = \frac{P(X < t)}{P(X < 6)} = \frac{\frac{t-4}{6}}{1/3} = \frac{t-4}{2}.$$

3.25 (a) If f is a pdf then $\int_{-\infty}^{\infty} f(x)dx = 1$. We have

$$1 = \int_{-\infty}^{\infty} f(x)dx = \int_1^3 (x^2 - b)dx = x^3/3 - bx \Big|_{x=1}^{x=3} = \frac{26}{3} - 2b.$$

This gives $b = \frac{23}{6}$. However, $x^2 - \frac{23}{6}$ is negative for $1 \leq x < \sqrt{\frac{23}{6}} \approx 1.96$ which shows that there is no b that would make the function f a pdf.

(b) We need $b \geq 0$, otherwise the function is zero everywhere. The $\cos x$ function is non-negative on $[-\pi/2, \pi/2]$, but then it goes below 0. Thus if g is a pdf then $b \leq \pi/2$. Computing the integral of g on $(-\infty, \infty)$ we get

$$\int_{-\infty}^{\infty} g(x)dx = \int_{-b}^b \cos(x)dx = 2 \sin(b).$$

There is exactly one solution for $2 \sin(b) = 1$ in the interval $(0, \pi/2]$, this is $b = \arcsin(1/2) = \pi/6$. For this choice of b the function g is a pdf.

3.26 (a) We require that the probability mass function sum to one. Hence,

$$1 = \sum_{k=1}^{\infty} p_X(k) = \sum_{k=1}^{\infty} \frac{c}{k(k+1)}.$$

The sum can be computed in the following way:

$$\begin{aligned} \sum_{k=1}^{\infty} \frac{c}{k(k+1)} &= \lim_{M \rightarrow \infty} \sum_{k=1}^M \frac{c}{k(k+1)} = c \lim_{M \rightarrow \infty} \sum_{k=1}^M \left(\frac{1}{k} - \frac{1}{k+1} \right) \\ &= c \lim_{M \rightarrow \infty} \left(1 - \frac{1}{2} + \frac{1}{2} - \frac{1}{3} + \frac{1}{3} - \frac{1}{4} + \cdots + \frac{1}{M} - \frac{1}{M+1} \right) \\ &= c \lim_{M \rightarrow \infty} \left(1 - \frac{1}{M+1} \right) = c. \end{aligned}$$

Combining the above shows that $c = 1$.

3.31 (a) We must have $1 = \int_{-\infty}^{\infty} f(x)dx$. So, we solve:

$$1 = \int_1^{\infty} cx^{-4}dx = \frac{c}{3}$$

which gives $c = 3$. For this value $f \geq 0$ everywhere, so we get a PDF.

(b) f is 0 on $[0.5, 1]$, so we have

$$P(0.5 < X < 1) = \int_{0.5}^1 f(x)dx = \int_{0.5}^1 0dx = 0.$$

(c) We have

$$P(0.5 < X < 2) = \int_{0.5}^2 f(x)dx = \int_1^2 3x^{-4}dx = -x^{-3} \Big|_{x=1}^2 = 1 - \frac{1}{8} = \frac{7}{8},$$

where we used that $f(x)$ is zero on $[0.5, 1]$, and it is $3x^{-4}$ on $[1, 2]$.

(d) We have

$$P(2 < X < 4) = \int_2^4 f(x)dx = \int_2^4 3x^{-4}dx = -x^{-3} \Big|_{x=2}^4 = \frac{1}{8} - \frac{1}{64} = \frac{7}{64}.$$

Homework 5 – Solutions

- 3.5 The possible values of a discrete random variable are exactly the values where the c.d.f. jumps. In this case these are the values 1, $4/3$, $3/2$ and $9/5$. The corresponding probabilities are equal to the size of corresponding jumps:

$$\begin{aligned} p_X(1) &= \frac{1}{3} - 0 = \frac{1}{3}, \\ p_X(4/3) &= \frac{1}{2} - \frac{1}{3} = \frac{1}{6}, \\ p_X(3/2) &= \frac{3}{4} - \frac{1}{2} = \frac{1}{4}, \\ p_X(9/5) &= 1 - \frac{3}{4} = \frac{1}{4}. \end{aligned}$$

- 3.7 (a) For a continuous random variable

$$P(a \leq X \leq b) = P(a < X \leq b) = F(b) - F(a)$$

(since $P(X = a) = 0$). The CDF F is non-decreasing, and its values are in $[0, 1]$, so $P(a \leq X \leq b) = F(b) - F(a) = 1$ exactly if $F(b) = 1$ and $F(a) = 0$. From the definition of F we see that the largest possible value of a where $F(a) = 0$ is $a = \sqrt{2}$, and the smallest possible value of b where $F(b) = 1$ is $b = \sqrt{3}$. Hence $a = \sqrt{2}$ and $b = \sqrt{3}$ gives the smallest such interval.

(b) Since X is continuous, $P(X = 1.6) = 0$. We can also see this directly from F :

$$P(X = 1.6) = F(1.6) - \lim_{x \rightarrow 1.6^-} F(x).$$

Since $F(x)$ is continuous at $x = 1.6$ (actually, it is continuous everywhere), we have $\lim_{x \rightarrow 1.6^-} F(1.6) = F(1.6)$ and this gives $P(X = 1.6) = 0$ again.

(c) Because X is continuous, we have $P(1 \leq X \leq 3/2) = P(1 < X \leq 3/2)$. We also have

$$\begin{aligned} P(1 \leq X \leq 3/2) &= P(1 < X \leq 3/2) = P(X \leq 3/2) - P(X \leq 1) \\ &= F(3/2) - F(1) = ((\frac{3}{2})^2 - 2) - 0 = \frac{9}{4} - 2 = \frac{1}{4}. \end{aligned}$$

We used $1 < \sqrt{2} \leq 3/2 \leq \sqrt{3}$ when we evaluated $F(3/2) - F(1)$.

(d) Since F is continuous, and it is differentiable apart from finitely many points ($\sqrt{2}$ and $\sqrt{3}$), we can just differentiate it to get the probability density function:

$$f(x) = F'(x) = \begin{cases} 2x & \text{if } \sqrt{2} < x < \sqrt{3} \\ 0 & \text{otherwise.} \end{cases}$$

We chose 0 for the value of f at $\sqrt{2}$ and $\sqrt{3}$, but the actual values are not important.

- 3.9 (a) Since X is continuous, we can compute its mean as

$$E[X] = \int_{-\infty}^{\infty} x f(x) dx = \int_0^{\infty} x \cdot 3e^{-3x} dx.$$

Using integration by parts we can evaluate the last integral to get $E[X] = \frac{1}{3}$.

(b) e^{2X} is a function of X , and X is continuous, so we can compute $E[e^{2X}]$ as follows:

$$E[e^{2X}] = \int_{-\infty}^{\infty} e^{2x} f(x) dx = \int_0^{\infty} e^{2x} \cdot 3e^{-3x} dx = \int_0^{\infty} 3e^{-x} dx = 3.$$

3.10 (a) The random variable $|X|$ takes values 0 and 1 with probabilities

$$P(|X| = 0) = P(X = 0) = \frac{1}{3} \quad \text{and} \quad P(|X| = 1) = P(X = 1) + P(X = -1) = \frac{2}{3}.$$

Then the definition of expectation gives

$$E[|X|] = 0 \cdot P(|X| = 0) + 1 \cdot P(|X| = 1) = \frac{2}{3}.$$

(b) We have

$$\begin{aligned} E[|X|] &= \sum_k |k| P(X = k) = 1 \cdot P(X = -1) + 0 \cdot P(X = 0) + 1 \cdot P(X = 1) \\ &= \frac{1}{2} + \frac{1}{6} = \frac{2}{3}. \end{aligned}$$

3.47 (a) We need to find $F(x) = P(X \leq x)$ for all x . The X coordinate of a point in the triangle must be between 0 and 30, so $F(x) = 0$ for $x < 0$ and $F(x) = 1$ for $x \geq 30$.

For $0 \leq x < 30$ then the set of points in the triangle with $X \leq x$ is the triangle with vertices $(0, 0)$, $(x, 0)$ and $(x, \frac{2}{3}x)$. The area of this triangle is $\frac{1}{3}x^2$, while the area of the original triangle is $\frac{20 \cdot 30}{2} = 300$. This means that if $0 \leq x < 30$ then $F(x) = \frac{\frac{1}{3}x^2}{300} = \frac{x^2}{900}$. Thus

$$F(x) = \begin{cases} 0 & x < 0 \\ \frac{x^2}{900} & 0 \leq x < 30 \\ 1 & x \geq 30 \end{cases}.$$

(b) Since $F(x)$ is continuous everywhere, and it is differentiable everywhere apart from $x = 30$ we can get the probability density function as $F'(x)$. This gives

$$f(x) = \begin{cases} \frac{x}{450} & 0 \leq x < 30 \\ 0 & \text{otherwise} \end{cases}.$$

(c) Since X is a continuous random variable, we can compute $E[X]$ as

$$E[X] = \int_{-\infty}^{\infty} xf(x)dx.$$

Using the solution from part (b):

$$E[X] = \int_{-\infty}^{\infty} xf(x)dx = \int_0^{30} x \frac{x}{450} dx = 20.$$

3.48 Denote the distance by R . The distance from the y -axis for a point in the triangle is at most 2, hence $0 \leq R \leq 2$. We first compute the c.d.f. of R . For $0 < r < 2$ the event $\{R < r\}$ is the same as the event that the chosen point is in the trapezoid with vertices $(0, 0)$, $(r, 0)$, $(r, 1 - r/2)$, $(0, 1)$. The probability of this event can be computed by taking ratios of areas:

$$F_R(r) = P(R \leq r) = \frac{\text{area(trapezoid)}}{\text{area(triangle)}} = \frac{\frac{r(1+1-r/2)}{2}}{\frac{2 \cdot 1}{2}} = r - \frac{r^2}{4}.$$

For $r \geq 2$ we have $F_R(r) = P(R \leq r) = 1$ and for $r \leq 0$ we have $F_R(r) = P(R \leq r) = 0$. The found c.d.f. is continuous everywhere and differentiable apart from $r = 0$. Thus R is continuous and we can find the probability density function by differentiation:

$$f_R(r) = (F_R(r))' = 1 - r/2, \quad \text{if } 0 < r < 2,$$

and $f_R(r) = 0$ otherwise.

We can compute the expectation of R by evaluating the appropriate integral:

$$E[R] = \int_{-\infty}^{\infty} r f_R(R) dr = \int_0^2 r(1 - r/2) dr = \frac{2}{3}.$$

Homework 6 – Solutions

3.16 Since Z is continuous, we can compute its expectation as

$$E[Z] = \int_{-\infty}^{\infty} z f_Z(z) dz = \int_1^2 z \cdot \frac{1}{7} dz + \int_5^7 z \cdot \frac{3}{7} dz = \frac{1}{7} \cdot \frac{4-1}{2} + \frac{3}{7} \cdot \frac{49-25}{2} = \frac{75}{14}.$$

The second moment is

$$\begin{aligned} E[Z^2] &= \int_{-\infty}^{\infty} z^2 f_Z(z) dz = \int_1^2 z^2 \cdot \frac{1}{7} dz + \int_5^7 z^2 \cdot \frac{3}{7} dz \\ &= \frac{1}{7} \cdot \frac{8-1}{3} + \frac{3}{7} \cdot \frac{7^3-5^3}{3} = \frac{661}{21}. \end{aligned}$$

Hence, the variance is

$$\text{Var}(Z) = E[Z^2] - (E[Z])^2 = \frac{661}{21} - \left(\frac{75}{14}\right)^2 = \frac{1633}{588}.$$

3.18 If $X \sim \mathcal{N}(\mu, \sigma^2)$ then $Z = \frac{X-\mu}{\sigma}$ is a standard normal random variable. Recall that the values $P(Z \leq x) = P(Z < x) = \Phi(x)$ can be looked up using the table in Appendix E.

(a)

$$\begin{aligned} P(2 < X < 6) &= P\left(\frac{2-3}{2} < \frac{X-3}{2} < \frac{6-3}{2}\right) = P(-\tfrac{1}{2} < Z < \tfrac{3}{2}) \\ &= P(Z < 1.5) - P(Z < -.5) = \Phi(1.5) - \Phi(-0.5) \\ &= \Phi(1.5) - (1 - \Phi(0.5)) = 0.9332 - (1 - 0.6915) = .6247. \end{aligned}$$

(b) We need c so that

$$0.33 = P(X > c) = P\left(\frac{X-3}{2} > \frac{c-3}{2}\right) = 1 - \Phi\left(\frac{c-3}{2}\right).$$

Hence, we need c satisfying $\Phi\left(\frac{c-3}{2}\right) = 0.67$. Checking the table in Appendix E, we conclude that $\Phi(z) = 0.67$ is solved by $z = 0.44$. Hence,

$$\frac{c-3}{2} = 0.44 \iff c = 3.88.$$

(c) Using $\text{Var}(X) = E[X^2] - (E[X])^2$ we have that

$$E[X^2] = \text{Var}(X) + (E[X])^2 = 4 + 3^2 = 13.$$

3.23 (a) The possible values for the profit (in dollars) are $0 - 1 = -1$, $2 - 1 = 1$, $100 - 1 = 99$ and $7000 - 1 = 6999$. The probability mass function can be computed as follows:

$$P(X = -1) = P(\text{the randomly chosen player was not a winner}) = \frac{10000 - 100}{10000} = \frac{99}{100},$$

$$P(X = 1) = P(\text{the randomly chosen player was one of the 80 who won \$2}) = \frac{80}{10000} = \frac{1}{125},$$

$$P(X = 99) = P(\text{the randomly chosen player was one of the 19 who won \$100}) = \frac{19}{10000},$$

$$P(X = 6999) = P(\text{the randomly chosen player was the one who won \$7000}) = \frac{1}{10000}.$$

(b)

$$P(X \geq 100) = P(X = 6999) = \frac{1}{10000}.$$

(c) Since X is discrete, we can find its expectation as

$$E[X] = \sum_k kP(X = k) = -1 \cdot \frac{99}{100} + 1 \cdot \frac{1}{125} + 99 \cdot \frac{19}{10000} + 6999 \cdot \frac{1}{10000} = -0.094.$$

For the variance we need $E[X^2]$:

$$E[X^2] = \sum_k k^2 P(X = k) = 1 \cdot \frac{99}{100} + 1 \cdot \frac{1}{125} + 99^2 \cdot \frac{19}{10000} + 6999^2 \cdot \frac{1}{10000} = 4918.22.$$

From this we get

$$\text{Var}(X) = E[X^2] - (E[X])^2 \approx 4918.21.$$

- 3.28 (a) The possible values of X are 1, 2, and 3. (If we don't get a prize in the first two boxes, then the remaining three will all have prizes in them, so we will definitely get a prize for the third box.)

Since there are three boxes with nice prizes, we have

$$P(X = 1) = \frac{3}{5}.$$

(Out of the five equally likely outcomes three will give us a prize.)

To compute $P(X = 2)$ note that $\{X = 2\}$ means that we took a sample of two boxes (without replacement) in order, the first had no prize, and the second had a prize. We have $5 \cdot 4$ equally likely outcomes, and out of these $2 \cdot 3$ will satisfy the conditions. (2 choices for the box with no nice prize, and 3 choices for the box with a nice prize.) Hence,

$$P(X = 2) = \frac{2 \cdot 3}{5 \cdot 4} = \frac{3}{10}.$$

Similarly,

$$P(X = 3) = \frac{2 \cdot 1 \cdot 3}{5 \cdot 4 \cdot 3} = \frac{1}{10}.$$

We could also get this by taking the complement of $\{X = 3\}$ to get

$$P(X = 3) = 1 - P(X = 1) - P(X = 2) = 1 - \frac{3}{5} - \frac{3}{10} = \frac{1}{10}.$$

- (b) This is a discrete random variable, we can compute the expectation as

$$E[X] = 1 \cdot \frac{3}{5} + 2 \cdot \frac{3}{10} + 3 \cdot \frac{1}{10} = \frac{3}{2}.$$

- (c) The second moment is

$$E[X^2] = 1^2 \cdot \frac{3}{5} + 2^2 \cdot \frac{3}{10} + 3^2 \cdot \frac{1}{10} = \frac{27}{10}.$$

Hence, the variance is

$$\text{Var}(X) = E[X^2] - (E[X])^2 = \frac{27}{10} - \left(\frac{3}{2}\right)^2 = \frac{9}{20}.$$

(d) Let W be the gain or loss in this game. Then

$$W = 100(2 - X) = 200 - 100X = \begin{cases} 100, & \text{if } X = 1 \\ 0, & \text{if } X = 2 \\ -100, & \text{if } X = 3. \end{cases}$$

We can use $E[200 - 100X] = 200 - 100E[X]$ to finish the computation:

$$E[W] = E[200 - 100X] = 200 - 100E[X] = 200 - 100 \cdot \frac{3}{2} = 50.$$

3.66 We can express X as $X = \sqrt{3}Y + 8$ where $Y \sim \mathcal{N}(0, 1)$. Then

$$0.15 = P(X > \alpha) = P(\sqrt{3}Y + 8 > \alpha) = P(Y > \frac{\alpha-8}{\sqrt{3}}) = 1 - \Phi(\frac{\alpha-8}{\sqrt{3}}).$$

Using the table in Appendix E we get that if $\Phi(\frac{\alpha-8}{\sqrt{3}}) = 0.85$ then $\frac{\alpha-8}{\sqrt{3}} \approx 1.04$. From this we get

$$\alpha \approx \sqrt{3}1.04 + 8 \approx 9.8.$$

3.71 We define noon to be time zero. Let $X \sim N(0, 36)$ model the arrival time of the bus in minutes (since the standard deviation is 6). Thus, $X = 6Z$ where $Z \sim N(0, 1)$. From this we get

$$P(X > 5) = P(6Z > 5) = P(Z > 5/6) = 1 - \Phi(0.83) \approx 1 - 0.7967 = 0.2033.$$

Homework 7 – Solutions

- 4.4 Let S_n be the number of times the roll is 3, 4, 5 or 6 in the first n rolls. Then $X_n = 2S_n + (n - S_n) = S_n + n$ and $S_n \sim \text{Bin}(n, \frac{2}{3})$. We have $E(S_{90}) = 60$ and $\text{Var}(S_{90}) = 90 \cdot \frac{2}{3} \cdot \frac{1}{3} = 20$. Since the variance is larger than 10 we can safely use the normal approximation.

For this we first rewrite the probability $P(X_{90} \geq 160)$ in terms of the standardized version of S_{90} and then use the normal approximation:

$$\begin{aligned} P(X_{90} \geq 160) &= P(S_{90} \geq 70) = P\left(\frac{S_{90} - 60}{\sqrt{20}} \geq \frac{70 - 60}{\sqrt{20}}\right) = P\left(\frac{S_{90} - 60}{\sqrt{20}} \geq \sqrt{5}\right) \\ &\approx 1 - \Phi(2.24) \approx 1 - 0.9875 = 0.0125. \end{aligned}$$

- 4.6 Let n be the size of the sample and S_n the number of positive answers in the sample. Then $\hat{p} = \frac{S_n}{n}$ and we need $P(|\hat{p} - p| \leq 0.02) \geq 0.95$.

We have seen that $P(|\hat{p} - p| > \varepsilon)$ can be estimated using the normal approximation as

$$P(|\hat{p} - p| < \varepsilon) \approx 2\Phi\left(\frac{\varepsilon\sqrt{n}}{\sqrt{p(1-p)}}\right) - 1 \geq 2\Phi(2\varepsilon\sqrt{n}) - 1.$$

Here we have $\varepsilon = 0.02$ and need $2\Phi(2\varepsilon\sqrt{n}) - 1 \geq 0.95$. This leads to $\Phi(2\varepsilon\sqrt{n}) \geq 0.975$ which, by the table of Φ -values, is satisfied if $2\varepsilon\sqrt{n} \geq 1.96$. Solving this inequality gives

$$n \geq \frac{1.96^2}{4\varepsilon^2} = 2401.$$

Thus the size of the sample should be at least 2401.

- 4.8 We have $n = 1,000,000$ trials with an unknown success probability p . To find a 99.9% confidence interval we need an $\varepsilon > 0$ so that $P(|\hat{p} - p| < \varepsilon) \geq 0.999$, where $\hat{p}$ is the fraction of positive outcomes. We have seen that $P(|\hat{p} - p| < \varepsilon)$ can be estimated using the normal approximation as

$$P(|\hat{p} - p| < \varepsilon) \approx 2\Phi\left(\frac{\varepsilon\sqrt{n}}{\sqrt{p(1-p)}}\right) - 1 \geq 2\Phi(2\varepsilon\sqrt{n}) - 1.$$

We need $2\Phi(2\varepsilon\sqrt{n}) - 1 \geq 0.999$ which means $\Phi(2\varepsilon\sqrt{n}) \geq 0.9995$ and so approximately $2\varepsilon\sqrt{n} \geq 3.32$. (Since 0.9995 appears several times in our table, other values instead of 3.32 are also acceptable.) This gives

$$\varepsilon \geq \frac{3.32}{2\sqrt{n}} \approx 0.00166$$

with $n = 1,000,000$. We had 180,000 positive outcomes, so $\hat{p} = 0.18$. Thus our confidence interval is $(0.18 - 0.00166, 0.18 + 0.00166) = (0.17834, 0.18166)$.

If we choose 3.28 from the table for the solution of $\Phi(x) = 0.9995$ then we get $(0.17836, 0.18164)$ instead.

- 4.10 It is reasonable to assume that the hockey player has a number of scoring chances per game, but only a few of them result in goals. Hence the number of goals in a given game corresponds to counting rare events, which means that it is reasonable to approximate this random number

with a $\text{Poisson}(\lambda)$ distributed random variable. Then the probability of scoring at least one goal would be $1 - e^{-\lambda}$ (since $e^{-\lambda}$ is the probability of no goals). Using the setup of the problem we have $1 - e^{-\lambda} \approx 0.5$ which gives $\lambda \approx \ln(2) \approx 0.6931$. We estimate the probability that the player scores exactly 3 goals. Using the Poisson probability mass function and our estimate on λ gives

$$P(\text{exactly 3 goals}) = \frac{\lambda^3}{3!} e^{-\lambda} \approx 0.028.$$

Thus we would expect the player to get a hat-trick in about 2.8% of his games.

One could also interpret the question the following way: what is the probability of scoring *at least* 3 goals in a game? In that case we get

$$\begin{aligned} P(\text{at least 3 goals}) &= 1 - P(\text{at most 2 goals}) = 1 - e^{-\lambda} - \lambda e^{-\lambda} - \frac{\lambda^2}{2!} e^{-\lambda} \\ &= 1 - \frac{1}{2} (1 + \ln 2 + \frac{1}{2} (\ln 2)^2) \approx 0.033. \end{aligned}$$

Both calculations give the answer of roughly 3 percent.

- 4.16 (a) Denote by S the number of random numbers starting with the digit 1. Note that a number in the interval $[1.5, 4.8]$ starts with 1 if and only if it is in the interval $[1.5, 2)$. The probability that a uniformly chosen number from the interval $[1.5, 4.8]$ is in $[1.5, 2)$ is equal to $p = \frac{0.5}{4.8-1.5} = \frac{5}{33}$. Assuming that the 500 numbers are chosen independently, the distribution of S is binomial with parameters $n = 500$ and p .

To estimate $P(S < 65)$ we use normal approximation. Note that $E[S] = np = 500 \cdot \frac{5}{33} \approx 75.7576$ and $\text{Var}(S) = np(1-p) \approx 64.2792$. Hence

$$\begin{aligned} P(S < 65) &= P\left(\frac{S - 75.7576}{\sqrt{64.2792}} < \frac{65 - 75.7576}{\sqrt{64.2792}}\right) \approx P\left(\frac{S - 75.7576}{\sqrt{64.2792}} < -1.34\right) \\ &\approx \Phi(-1.34) = 1 - \Phi(1.34) \approx 1 - 0.9099 = 0.0901. \end{aligned}$$

Note that $P(S < 65) = P(S \leq 64)$. Using 64 instead of 65 in the calculation above gives $1 - \Phi(1.47) \approx 0.0708$. If we use the continuity correction then we need to use 64.5 instead of 65 which gives $1 - \Phi(1.4) \approx 0.0808$. The actual probability (evaluated numerically) is 0.0778.

- (b) We proceed similarly as in part (a). The probability that a given uniformly chosen number from $[1.5, 4.8]$ starts with 3 is $q = \frac{1}{3.3} = \frac{10}{33}$. If we denote the number of such numbers among the 500 random numbers by T then $T \sim \text{Bin}(n, q)$ with $n = 500$.

Then

$$\begin{aligned} P(T > 160) &= P\left(\frac{T - nq}{\sqrt{nq(1-q)}} > \frac{160 - nq}{\sqrt{nq(1-q)}}\right) \approx P\left(\frac{T - nq}{\sqrt{nq(1-q)}} > 0.83\right) \\ &\approx 1 - \Phi(0.83) \approx 1 - 0.7967 = 0.2033. \end{aligned}$$

Again, since $P(T > 160) = P(T \geq 161)$, we could have done the computation with 161 instead of 160, which would give $1 - \Phi(0.92) \approx 0.1788$. If we use the continuity correction then we replace 160 with 160.5 in the calculation above which leads to $1 - \Phi(0.87) \approx 0.1922$. The actual probability (evaluated numerically) is 0.1906.

4.26 Let S be the number of interviewed people that prefer whole milk to skim milk. Then $S \sim \text{Bin}(n, p)$ with $n = 100$. Our estimate for p is $\hat{p} = \frac{S}{n}$. The event $p \in (\hat{p} - 0.1, \hat{p} + 0.1)$ is the same as $|\frac{S}{n} - p| < 0.1$. To estimate the probability of this event we use normal approximation:

$$\begin{aligned} P(|S/n - p| < 0.1) &= P\left(-\frac{0.1\sqrt{n}}{\sqrt{p(1-p)}} < \frac{S-np}{\sqrt{np(1-p)}} < \frac{0.1\sqrt{n}}{\sqrt{p(1-p)}}\right) \\ &\approx 2\Phi\left(\frac{0.1\sqrt{n}}{\sqrt{p(1-p)}}\right) - 1 \geq 2\Phi(0.2\sqrt{n}) - 1, \end{aligned}$$

where we used $p(1-p) \leq 1/4$ in the last step.

Since $n = 100$ we have

$$2\Phi(0.2\sqrt{n}) - 1 = 2\Phi(2) - 1 \approx 2 \cdot 0.9772 - 1 = 0.9544.$$

Thus the interval $(\hat{p} - 0.1, \hat{p} + 0.1)$ corresponds to 95.44% confidence.

Homework 8 – Solutions

- 4.34 We can assume that each taxi has a small probability of getting into an accident on a given day, independently of the others. Since there are a large number of taxis, the number of accidents on a given week could be well approximated with a Poisson(μ) distributed random variable. There are on average 3 accidents a week, thus it is reasonable to choose $\mu = 3$. Then the probability of having at most 2 accidents next week is given by

$$e^{-3} \left(\frac{3^0}{0!} + \frac{3^1}{1!} + \frac{3^2}{2!} \right) = e^{-3} \left(1 + 3 + \frac{9}{2} \right).$$

- 4.42 (a) Let X be the number of mildly defective gadgets in the box. Then $X \sim \text{Bin}(n, p)$ with $n = 100$ and $p = 0.2 = \frac{1}{5}$. We have

$$P(A) = P(X < 15) = \sum_{k=0}^{14} \binom{100}{k} (1/5)^k (4/5)^{100-k}.$$

- (b) We have $np(1-p) = 16 > 10$ and $np^2 = 4$. This suggests that the normal approximation is more appropriate than the Poisson approximation in this case.

Using normal approximation we get

$$\begin{aligned} P(X < 15) &= P\left(\frac{X - 100 \cdot \frac{1}{5}}{\sqrt{100 \cdot \frac{1}{5} \cdot \frac{4}{5}}} < \frac{15 - 100 \cdot \frac{1}{5}}{\sqrt{100 \cdot \frac{1}{5} \cdot \frac{4}{5}}}\right) \\ &= P\left(\frac{X - 100 \cdot \frac{1}{5}}{\sqrt{100 \cdot \frac{1}{5} \cdot \frac{4}{5}}} < -\frac{5}{4}\right) \\ &\approx \Phi(-1.25) = 1 - \Phi(1.25) \approx 1 - 0.8944 = 0.1056. \end{aligned}$$

With continuity correction we would get $\Phi(-1.375) = 1 - \Phi(1.375) \approx 0.08455$ (using linear interpolation to get $\Phi(1.375)$).

- 4.46 The probability that we get 5 tails when we flip a coin 5 times is $\frac{1}{2^5} = \frac{1}{32}$. Thus $X \sim \text{Bin}(n, p)$ with $n = 30$ and $p = \frac{1}{32}$. Since $np(1-p) = \frac{465}{512} < 1$, the normal approximation is not appropriate. On the other hand, $np^2 = \frac{15}{512} \approx 0.029$ is small, so the Poisson approximation should work. For this we approximate the distribution of X using a random variable $Y \sim \text{Poisson}(\lambda)$ with $\lambda = np = \frac{15}{16}$ to get

$$P(X = 2) \approx P(Y = 2) = \frac{\lambda^2}{2} e^{-\lambda} = \left(\frac{15}{16}\right)^2 \left(\frac{1}{2}\right) e^{-\frac{15}{16}} \approx 0.1721.$$

- 5.2 (a) The moment generating function finite near zero (in fact: everywhere), so we can find the first and second moment by looking at the first and second derivative at 0.

We have

$$M'(t) = -\frac{4}{3}e^{-4t} + \frac{5}{6}e^{5t}, \quad M''(t) = \frac{16}{3}e^{-4t} + \frac{25}{6}e^{5t}.$$

Hence $E(X) = M'(0) = -\frac{4}{3} + \frac{5}{6} = -\frac{1}{2}$, $E(X^2) = M''(0) = \frac{16}{3} + \frac{25}{6} = \frac{57}{6} = \frac{19}{2}$, and $\text{Var}(X) = E(X^2) - (E[X])^2 = \frac{19}{2} - \frac{1}{4} = \frac{37}{4}$.

- (b) Since the moment generating function is finite near zero, it uniquely identifies the distribution. From the form of the moment generating function we see that X is discrete, the possible values are $-4, 0$ and 5 . The corresponding probabilities can be read off from the coefficients of the appropriate exponential terms:

$$p(0) = \frac{1}{2}, \quad p(-4) = \frac{1}{3}, \quad p(5) = \frac{1}{6}.$$

From this we get

$$\begin{aligned} E(X) &= \frac{1}{3} \cdot (-4) + \frac{1}{6} \cdot 5 = -\frac{1}{2}, \\ E(X^2) &= \frac{1}{3} \cdot 16 + \frac{1}{6} \cdot 25 = \frac{57}{6} = \frac{19}{2}, \\ \text{Var}(X) &= E(X^2) - (E[X])^2 = \frac{19}{2} - \frac{1}{4} = \frac{37}{4}, \end{aligned}$$

and these values agree with what we got in part (a).

- 5.8 We first compute the cumulative distribution function of Y . Since $-1 \leq X \leq 2$, we have $0 \leq X^2 \leq 4$, thus $F_Y(y) = 1$ for $y \geq 4$ and $F_Y(y) = 0$ for $y < 0$.

For $0 \leq y < 4$ we have

$$F_Y(y) = P(Y \leq y) = P(X^2 \leq y) = P(-\sqrt{y} \leq X \leq \sqrt{y}) = F_X(\sqrt{y}) - F_X(-\sqrt{y}).$$

Differentiating this we get the probability density function:

$$f_Y(y) = F'_Y(y) = \frac{1}{2\sqrt{y}} f_X(\sqrt{y}) + \frac{1}{2\sqrt{y}} f_X(-\sqrt{y}).$$

The probability density of X is $f_X(x) = \frac{1}{3}$ for $-1 \leq x \leq 2$ and zero otherwise. For $0 \leq y \leq 1$ then both $f_X(\sqrt{y})$ and $f_X(-\sqrt{y})$ is equal to $\frac{1}{3}$, and for $1 < y < 4$ we have $f_X(\sqrt{y}) = \frac{1}{3}$ and $f_X(-\sqrt{y}) = 0$.

From this we get

$$f_Y(y) = \begin{cases} \frac{1}{3\sqrt{y}} & \text{for } 0 \leq y \leq 1, \\ \frac{1}{6\sqrt{y}} & \text{for } 1 < y < 4, \\ 0 & \text{otherwise.} \end{cases}$$

- 5.10 Using the Binomial Theorem we get

$$M(t) = \left(\frac{1}{5} + \frac{4}{5} e^t \right)^{30} = \sum_{k=0}^{30} \binom{30}{k} \left(\frac{4}{5} \right)^k e^{kt} \left(\frac{1}{5} \right)^{30-k}.$$

Since this is the sum of terms of the form $p_k e^{tk}$, we see that X is discrete. The possible values can be identified with the exponents: these are $0, 1, 2, \dots, 30$. The coefficients are the corresponding probabilities:

$$P(X = k) = \binom{30}{k} \left(\frac{4}{5} \right)^k \left(\frac{1}{5} \right)^{30-k}, \quad k = 0, 1, \dots, 30.$$

We can recognize this as the probability mass function of a binomial distribution with $n = 30$ and $p = \frac{4}{5}$. Since the moment generating function identifies the distribution (if it is finite near $t = 0$), this shows that X must have $\text{Bin}(30, \frac{4}{5})$ distribution.

5.18 (a) Using the definition of a moment generating function we have

$$\begin{aligned} M_X(t) &= E[e^{tX}] = \sum_{k=1}^{\infty} e^{tk} P(X=k) = \sum_{k=1}^{\infty} (e^t)^k (1-p)^{k-1} p \\ &= pe^t \sum_{k=1}^{\infty} (e^t(1-p))^{k-1} = pe^t \sum_{k=0}^{\infty} (e^t(1-p))^k \end{aligned}$$

Note that the sum is just a geometric series, it converges to a finite number if and only if $|e^t(1-p)| = e^t(1-p) < 1$, which holds if and only if $t < \ln\left(\frac{1}{1-p}\right)$. In this case we have

$$M_X(t) = pe^t \cdot \frac{1}{1 - e^t(1-p)}.$$

Overall, we find:

$$M_X(t) = \begin{cases} \frac{pe^t}{1 - e^t(1-p)} & t < \ln\left(\frac{1}{1-p}\right) \\ \infty & t \geq \ln\left(\frac{1}{1-p}\right) \end{cases}.$$

(b) For the mean,

$$\begin{aligned} E[X] &= M'_X(0) = \left. \frac{pe^t(1 - e^t(1-p)) - pe^t(-e^t(1-p))}{(1 - e^t(1-p))^2} \right|_{t=0} \\ &= \left. \frac{pe^t}{(1 - e^t(1-p))^2} \right|_{t=0} \\ &= \frac{p}{p^2} = \frac{1}{p}. \end{aligned}$$

For the variance we need the second moment,

$$\begin{aligned} E[X^2] &= M''_X(0) = \left. \frac{pe^t(1 - e^t(1-p))^2 - 2pe^t(1 - e^t(1-p))(-e^t(1-p))}{(1 - e^t(1-p))^4} \right|_{t=0} \\ &= \frac{p(1 - (1-p))^2 - 2p(1 - (1-p))(-(1-p))}{p^4} \\ &= \frac{p^3 - 2p^3 + 2p^2}{p^4} = \frac{2}{p^2} - \frac{1}{p}. \end{aligned}$$

Finally the variance is

$$\text{Var}(X) = E[X^2] - (E[X])^2 = \frac{2}{p^2} - \frac{1}{p} - \frac{1}{p^2} = \frac{1}{p^2} - \frac{1}{p}.$$

5.20 (a) From the definition we have

$$M_X(t) = \int_{-\infty}^{\infty} e^{tx} \frac{1}{2} e^{-|x|} dx = \frac{1}{2} \int_0^{\infty} e^{-(1-t)x} dx + \frac{1}{2} \int_{-\infty}^0 e^{(t+1)x} dx.$$

After the change of variables $x \rightarrow -x$ for the integral on $(-\infty, 0]$ we get

$$M_X(t) = \frac{1}{2} \int_0^{\infty} e^{-(1-t)x} dx + \frac{1}{2} \int_0^{\infty} e^{-(t+1)x} dx.$$

We have seen that the integral of $\int_0^\infty e^{-cx} dx$ is $\frac{1}{c}$ if $c > 0$ and ∞ otherwise. (E.g. by computing the integral of the PDF of an $\text{Exp}(c)$ distribution on the whole line.) $M_X(t)$ is finite exactly if both $\int_0^\infty e^{-(1-t)x} dx$ and $\int_0^\infty e^{-(t+1)x} dx$ are finite. Thus $M_X(t)$ is finite if $1-t > 0$ and $1+t > 0$ (or $-1 < t < 1$) and ∞ otherwise. Moreover, if it is finite then it is equal to

$$M_X(t) = \frac{1}{2} \cdot \frac{1}{1-t} + \frac{1}{2} \cdot \frac{1}{1+t} = \frac{1}{(1-t^2)}.$$

Thus $M_X(t)$ is $\frac{1}{(1-t^2)}$ for $|t| < 1$, and ∞ otherwise.

(b) We could try to differentiate $M_X(t)$ to get the moments, but it is simpler to take the Taylor expansion at $t = 0$. If $|t| < 1$ then $\frac{1}{1-t^2} = \sum_{k=0}^\infty t^{2k}$, hence

$$M_X(t) = \sum_{k=0}^\infty t^{2k}.$$

(We could also get the same answer by taking the expansions $\frac{1}{1-t} = \sum_{k=0}^\infty t^k$ and $\frac{1}{1+t} = \sum_{k=0}^\infty (-t)^k$, and take their average.)

The n th moment is the coefficient of t^n multiplied by $n!$. There are no odd exponent terms in the expansion, so all odd moments of X are zero. The term t^{2k} has a coefficient 1, so the $(2k)$ th moment is $(2k)!$.

5.24 (a) Since $Y = e^X > 0$, we have $F_Y(t) = 0$ for $t \leq 0$. Indeed, for $t \leq 0$,

$$F_Y(t) = P(Y \leq t) = P(e^X \leq t) = 0,$$

since $e^x > 0$ for all $x \in \mathbb{R}$. Next, for any $t > 0$

$$F_Y(t) = P(Y \leq t) = P(e^X \leq t) = P(X \leq \ln t) = \Phi(\ln t).$$

Differentiating this gives the probability density function for $t > 0$:

$$f_Y(t) = \Phi'(\ln t) \frac{1}{t} = \frac{1}{t} \varphi(\ln t) = \frac{1}{\sqrt{2\pi t^2}} \exp\left(-\frac{(\ln(t))^2}{2}\right).$$

For $t \leq 0$ the probability density function is 0.

(b) From the definition of Y we get that $E[Y^n] = E[(e^X)^n] = E[e^{nX}]$. Note that $E[e^{nX}] = M_X(n)$ is the moment generating function of X evaluated at n .

We computed the moment generating function for $X \sim \mathcal{N}(0, 1)$ and it is given by $M_X(t) = e^{t^2/2}$. Thus we have

$$E[Y^n] = e^{\frac{n^2}{2}}.$$

5.32 The set of possible values of X is $(0, 1)$, hence the set of possible values for Y is the interval $[1, \infty)$. Thus, for $t < 1$, $f_Y(t) = 0$. For $t \geq 1$,

$$P(Y \leq t) = P\left(\frac{1}{X} \leq t\right) = P\left(X \geq \frac{1}{t}\right) = 1 - \frac{1}{t}.$$

Differentiating now shows that $f_Y(t) = \frac{1}{t^2}$ when $t \geq 1$.

Homework 9 – Solutions

6.2 (a) The marginal probability mass function of X is found by computing the row sums,

$$P(X = 1) = \frac{1}{3}, \quad P(X = 2) = \frac{1}{2}, \quad P(X = 3) = \frac{1}{6}.$$

Computing the column sums gives the probability mass function of Y ,

$$P(Y = 0) = \frac{1}{5}, \quad P(Y = 1) = \frac{1}{5}, \quad P(Y = 2) = \frac{1}{3}, \quad P(Y = 3) = \frac{4}{15}.$$

(b) First we find the pairs of values of X and Y where $X + Y^2 \leq 2$. These are $(1, 0)$, $(1, 1)$, and $(2, 0)$. So we have

$$\begin{aligned} P(X + Y^2 \leq 2) &= P(X = 1, Y = 0) + P(X = 1, Y = 1) + P(X = 2, Y = 0) \\ &= \frac{1}{15} + \frac{1}{15} + \frac{1}{10} = \frac{7}{30}. \end{aligned}$$

6.6 (a) The marginal of X is

$$f_X(x) = \int_0^\infty x e^{-x(1+y)} dy = e^{-x} \int_0^\infty x e^{-xy} dy = e^{-x},$$

for $x > 0$ and zero otherwise. In the last step we can use that $g(y) = x e^{-xy}$ for $y > 0$, $g(y) = 0$ for $y \leq 0$ is the PDF of an exponential with parameter x , hence its integral on $[0, \infty)$ is equal to 1.

The marginal of Y is

$$f_Y(y) = \int_0^\infty x e^{-x(1+y)} dx = \frac{1}{(1+y)^2},$$

for $y > 0$ and zero otherwise. For this integral one can use integration by parts, or rewrite it as

$$\int_0^\infty x e^{-x(1+y)} dx = \frac{1}{1+y} \int_0^\infty x(1+y) e^{-x(1+y)} dx$$

and notice that the last integral is the expectation of an exponential with parameter $\lambda = 1 + y$, which is $\frac{1}{1+y}$.

(b) The expectation is

$$\begin{aligned} E[XY] &= \int_{-\infty}^\infty \int_{-\infty}^\infty xy f(x, y) dx dy = \int_0^\infty \int_0^\infty x^2 y e^{-x(1+y)} dy dx \\ &= \int_0^\infty x e^{-x} \int_0^\infty y x e^{-xy} dy dx = \int_0^\infty x e^{-x} \cdot \frac{1}{x} dx = \int_0^\infty e^{-x} dx = 1. \end{aligned}$$

In the third step we can use the fact that the y -integral is the expected value of an exponential with parameter x . In the last step we can use the fact that we are computing the integral of the PDF of an exponential with parameter 1.

(c) The expectation is

$$\begin{aligned} E\left[\frac{X}{1+Y}\right] &= \int_0^\infty \int_0^\infty \frac{x}{1+y} x e^{-x(1+y)} dx dy = \int_0^\infty \frac{1}{(1+y)^2} \int_0^\infty x^2 (1+y) e^{-x(1+y)} dx dy \\ &= \int_0^\infty \frac{1}{(1+y)^2} \cdot \frac{2}{(1+y)^2} dy = 2 \int_0^\infty \frac{1}{(1+y)^4} dy = \frac{2}{3}. \end{aligned}$$

We can notice that the integral $\int_0^\infty x^2 (1+y) e^{-x(1+y)} dx$ is the second moment of an exponential with parameter $\lambda = (1+y)$, which is equal to $\frac{2}{(1+y)^2}$. The last integral can be computed directly.

6.8 (a) X and Y from Exercise 6.2 are not independent. For example, note that $P(X = 3) > 0$ and $P(Y = 2) > 0$, but $P(X = 3, Y = 2) = 0$.

(b) The marginals for X and Y from Exercise 6.5 will be zero outside $[0, 1]$. Inside $[0, 1]$ they can be computed as follows:

For $0 \leq x \leq 1$,

$$f_X(x) = \int_{-\infty}^\infty f(x, y) dy = \frac{12}{7} \int_0^1 (xy + y^2) dy = \frac{12}{7} \left(\frac{1}{2}x + \frac{1}{3}\right) dy = \frac{6}{7}x + \frac{4}{7}.$$

For $0 \leq y \leq 1$,

$$f_Y(y) = \int_{-\infty}^\infty f(x, y) dx = \frac{12}{7} \int_0^1 (xy + y^2) dx = \frac{12}{7} \left(\frac{1}{2}y + y^2\right) dy = \frac{12}{7}y^2 + \frac{6}{7}y.$$

Thus, $f_X(x)f_Y(y) \neq f(x, y)$ as functions, which means that X and Y are not independent.

(d) X and Y are not independent. For example, choose any point (x, y) contained in the square $\{(u, v) : 0 \leq u \leq 1, 0 \leq v \leq 1\}$, but *not* contained in the triangle with vertices $(0, 0)$, $(1, 0)$, $(0, 1)$. Then $f_X(x) > 0$, $f_Y(y) > 0$, and so $f_X(x)f_Y(y) > 0$. However, $f(x, y) = 0$ (because the point is outside the triangle).

6.18 (a) The probability mass function can be visualized in tabular form

$X \backslash Y$	1	2	3	4
1	$\frac{1}{4}$	0	0	0
2	$\frac{1}{8}$	$\frac{1}{8}$	0	0
3	$\frac{1}{12}$	$\frac{1}{12}$	$\frac{1}{12}$	0
4	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{16}$

The terms are nonnegative and add to 1, which shows that $p_{X,Y}$ is a probability mass function.

(b) Adding the numbers in rows and columns gives the marginals PMFs. The marginal of X is

$$P(X = 1) = \frac{1}{4}, \quad P(X = 2) = \frac{1}{4}, \quad P(X = 3) = \frac{1}{4}, \quad P(X = 4) = \frac{1}{4},$$

whereas the marginal of Y is

$$P(Y = 1) = \frac{25}{48}, \quad P(Y = 2) = \frac{13}{48}, \quad P(Y = 3) = \frac{7}{48}, \quad P(Y = 4) = \frac{1}{16}.$$

(c) Collecting all pairs of (x, y) from the possible pairs of values with $x = y + 1$ we get

$$\begin{aligned} P(X = Y + 1) &= P(X = 2, Y = 1) + P(X = 3, Y = 2) + P(X = 4, Y = 3) \\ &= \frac{1}{8} + \frac{1}{12} + \frac{1}{16} = \frac{13}{48}. \end{aligned}$$

6.24 The number of green balls chosen is binomially distributed with parameters $n = 3$ and $p = \frac{1}{4}$. Hence, the probability that exactly two balls are green and one is not green is

$$\binom{3}{2} \left(\frac{1}{4}\right)^2 \frac{3}{4} = \frac{9}{64}.$$

The same argument goes for seeing exactly two red balls, two yellow balls, or two white balls. These events are disjoint, since we cannot have two different pairs with the same color within the 3 balls. Hence, the probability that exactly two balls are of the same color is

$$4 \cdot \frac{9}{64} = \frac{9}{16}.$$

We could also use the multinomial distribution: using G, R, Y, W for the number of green, red, yellow, white balls we see that the joint distribution of (G, R, Y, W) is multinomial with parameters $(3, 4, 1/4, 1/4, 1/4, 1/4)$. With nonnegative integers satisfying $g + r + y + w = 3$ we get

$$P(G = g, R = r, Y = y, W = w) = \binom{3}{g, r, y, w} (1/4)^3.$$

The event {exactly 2 balls are of the same color} is the disjoint union of the events $\{G = g, R = r, Y = y, W = w\}$ where g, r, y, w are 2, 1, 0, 0 in some order. Each one of these events has probability

$$\binom{3}{2, 1, 0, 0} (1/4)^3 = \frac{3!}{2!1!0!0!} (1/4)^3 = \frac{3}{64}$$

and there are 12 such events (we can choose the color which shows up twice 4 ways, and the other color that shows up 3 ways). This gives

$$P(\text{exactly 2 balls are of the same color}) = 12 \cdot \frac{3}{64} = \frac{36}{64} = \frac{9}{16}.$$

6.27 The possible values of Y are $-1, 1$, which is the same as X_2 . Thus, we need to show four things:

$$\begin{aligned} P(X_2 = 1, Y = 1) &= P(X_2 = 1)P(Y = 1) \\ P(X_2 = -1, Y = 1) &= P(X_2 = -1)P(Y = 1) \\ P(X_2 = 1, Y = -1) &= P(X_2 = 1)P(Y = -1) \\ P(X_2 = -1, Y = -1) &= P(X_2 = -1)P(Y = -1). \end{aligned}$$

To check the first one

$$P(X_2 = 1, Y = 1) = P(X_2 = 1, X_2 X_1 = 1) = P(X_2 = 1, X_1 = 1) = P(X_2 = 1)P(X_1 = 1) = \frac{p}{2}.$$

Also,

$$P(Y = 1) = P(X_1 = 1, X_2 = 1) + P(X_1 = -1, X_2 = -1) = \frac{p}{2} + \frac{1}{2} \cdot (1 - p) = \frac{1}{2},$$

and so,

$$P(X_2 = 1)P(Y = 1) = p\frac{1}{2} = P(X_2 = 1, Y = 1).$$

All the other terms are handled similarly, using $P(X_2 = a, Y = b) = P(X_1 = b/a, X_2 = a)$ and $P(Y = 1) = P(Y = -1) = 1/2$.

6.28 To help with notation we will use $q = 1 - p$.

For the joint probability mass function we need to compute $P(V = k, W = \ell)$ for all $k \geq 1, \ell = 0, 1, 2$. We have

$$\begin{aligned} P(V = k, W = 0) &= P(\min(X, Y) = k, X < Y) = P(X = k, k < Y) \\ &= P(X = k)P(k < Y) = pq^{k-1} \cdot q^k = pq^{2k-1}, \end{aligned}$$

where we used the independence of X and Y in the third equality.

We get $P(V = k, W = 2) = pq^{2k-1}$ in exactly the same way. Finally,

$$P(V = k, W = 1) = P(\min(X, Y) = k, X = Y) = P(X = k, Y = k) = p^2q^{2k-2}.$$

This gives us the joint probability mass function of V and W ; for the independence we need to check if this is the product of the marginals.

We have seen in class that we have $V \sim \text{Geom}(1 - q^2)$ so for any $k \in \{1, 2, \dots\}$ we get

$$P(V = k) = (1 - (1 - q^2))^{k-1}(1 - q^2) = q^{2k-2}(1 - q^2).$$

You can also get this from the joint probability mass function using

$$\begin{aligned} P(V = k) &= P(V = k, W = 0) + P(V = k, W = 1) + P(V = k, W = 2) \\ &= 2pq^{2k-1} + p^2q^{2k-2} = q^{2k-2}(2pq + p^2) = q^{2k-2}(1 - q^2) \end{aligned}$$

To compute the probability mass function of W we first use the independence of X and Y to get

$$\begin{aligned} P(W = 1) &= P(X = Y) = \sum_{k=1}^{\infty} P(X = k, Y = k) = \sum_{k=1}^{\infty} P(X = k)P(Y = k) \\ &= \sum_{k=1}^{\infty} pq^{k-1} \cdot pq^{k-1} = p^2 \sum_{k=0}^{\infty} (q^2)^k = \frac{p^2}{1 - q^2} \\ &= \frac{p}{2 - p}. \end{aligned}$$

To compute $P(W = 0)$ we can again use independence to get

$$\begin{aligned} P(W = 0) &= P(X < Y) = \sum_{k=1}^{\infty} P(X = k, Y > k) = \sum_{k=1}^{\infty} P(X = k)P(Y > k) \\ &= \sum_{k=1}^{\infty} pq^{k-1} \cdot q^k = pq \sum_{k=1}^{\infty} q^{2(k-1)} = \frac{pq}{1 - q^2} = \frac{1 - p}{2 - p}. \end{aligned}$$

A similar argument gives $P(W = 2) = \frac{1-p}{2-p}$. Note that these values can also be obtained using symmetry: we must have

$$P(W = 0) = P(X < Y) = P(Y < X) = P(W = 2),$$

and then

$$P(W = 0) = P(W = 2) = \frac{1}{2}(1 - P(W = 1)) = \frac{1-p}{2-p}.$$

Now we can check the independence of V and W . First note that

$$P(V = k)P(W = 0) = q^{2k-2}(1 - q^2)\frac{1-p}{2-p}, \quad P(V = k, W = 0) = pq^{2k-1},$$

and since $\frac{1-q^2}{2-p} = (1 - q)(1 + q)\frac{1}{1+q} = p$, we have

$$P(V = k)P(W = 0) = P(V = k, W = 0).$$

The same computation shows $P(V = k)P(W = 2) = P(V = k, W = 2)$. Finally,

$$P(V = k)P(W = 1) = q^{2k-2}(1 - q^2)\frac{p}{2-p}, \quad P(V = k, W = 1) = p^2q^{2k-2}$$

and using $\frac{1-q^2}{2-p} = p$ again we get

$$P(V = k)P(W = 1) = P(V = k, W = 1).$$

We showed that $P(V = k, W = \ell) = P(V = k)P(W = \ell)$ for all relevant k, ℓ , and this shows that V and W are independent.

6.30 We have $P(X = k) = p(1 - p)^{k-1}$ for $k = 1, 2, \dots$ and $P(Y = \ell) = \frac{\lambda^\ell}{\ell!}e^{-\lambda}$ for $\ell = 0, 1, 2, \dots$. Because of independence the joint PMF is given by

$$P(X = k, Y = \ell) = p(1 - p)^{k-1}\frac{\lambda^\ell}{\ell!}e^{-\lambda}, \quad k \geq 1, \ell \geq 0$$

with k, ℓ integers.

We need to compute $P(X - 1 = Y)$, we can do that by breaking the event up into smaller pieces based on the value of X , and using the joint PMF:

$$\begin{aligned} P(X - 1 = Y) &= \sum_{k=1}^{\infty} P(X = k, Y = k - 1) \\ &= \sum_{k=1}^{\infty} p(1 - p)^{k-1} \frac{\lambda^{k-1}}{(k-1)!} e^{-\lambda} = pe^{-\lambda} \sum_{k=1}^{\infty} \frac{(\lambda(1 - p))^{k-1}}{(k-1)!} \end{aligned}$$

We can rewrite the previous sum in terms of $n = k - 1$, with n going from 0 to ∞ :

$$P(X - 1 = Y) = pe^{-\lambda} \sum_{k=1}^{\infty} \frac{(\lambda(1 - p))^{k-1}}{(k-1)!} = pe^{-\lambda} \sum_{n=0}^{\infty} \frac{(\lambda(1 - p))^n}{n!} = pe^{-\lambda} \cdot e^{\lambda(1-p)}$$

where in the last step we evaluated the series using the series expansion of e^x .

Simplifying the result gives

$$P(X - 1 = Y) = pe^{-\lambda p}.$$

6.34 (a) The area of D is $\frac{3}{2}$, and hence the joint PDF is

$$f_{X,Y}(x,y) = \begin{cases} \frac{2}{3}, & (x,y) \in D \\ 0, & (x,y) \notin D. \end{cases}$$

The line segment from $(1,1)$ to $(2,0)$ that forms part of the boundary of D obeys the equation $y = 2 - x$. The marginal density functions are derived as follows. First for X .

For $x \leq 0$ and $x \geq 2$, $f_X(x) = 0$.

For $0 < x \leq 1$, $f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x,y) dy = \int_0^1 \frac{2}{3} dy = \frac{2}{3}$.

For $1 < x < 2$, $f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x,y) dy = \int_0^{2-x} \frac{2}{3} dy = \frac{4}{3} - \frac{2}{3}x$.

Let us check that this is a density function:

$$\int_{-\infty}^{\infty} f_X(x) dx = \int_0^1 \frac{2}{3} dx + \int_1^2 (\frac{4}{3} - \frac{2}{3}x) dx = 1,$$

so indeed it is.

Next the marginal density function of Y :

For $y \leq 0$ and $y \geq 1$, $f_Y(y) = 0$.

For $0 < y < 1$, $f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x,y) dx = \int_0^{2-y} \frac{2}{3} dx = \frac{4}{3} - \frac{2}{3}y$.

(b) Using the computed marginal densities we get

$$E[X] = \int_{-\infty}^{\infty} x f_X(x) dx = \int_0^1 \frac{2}{3}x dx + \int_1^2 (\frac{4}{3}x - \frac{2}{3}x^2) dx = \frac{7}{9}.$$

$$E[Y] = \int_{-\infty}^{\infty} y f_Y(y) dy = \int_0^1 (\frac{4}{3}y - \frac{2}{3}y^2) dy = \frac{4}{9}.$$

We could have also computed these with the following double integrals:

$$E[X] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x f(x,y) dx dy, \quad E[Y] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} y f(x,y) dx dy.$$

(c) X and Y are not independent. Their joint density is not a product of the marginal densities. Also, a picture of D shows that $P(X > \frac{3}{2}, Y > \frac{1}{2}) = 0$ because all points in D satisfy $x + y \leq 2$. However, the marginal densities show that $P(X > \frac{3}{2}) \cdot P(Y > \frac{1}{2}) > 0$ so the probability of the intersection does not equal the product of the probabilities.

6.36 (a) We need to find c so that $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x,y) dx dy = 1$. For this we need to compute

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-\frac{x^2}{2} - \frac{(x-y)^2}{2}} dx dy$$

We can decide whether we should integrate with respect to x or y first, and choosing y gives a slightly easier path.

$$\begin{aligned}\int_{-\infty}^{\infty} e^{-\frac{x^2}{2} - \frac{(x-y)^2}{2}} dy &= e^{-\frac{x^2}{2}} \int_{-\infty}^{\infty} e^{-\frac{(x-y)^2}{2}} dy \\ &= \sqrt{2\pi} e^{-\frac{x^2}{2}} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{(x-y)^2}{2}} dy = \sqrt{2\pi} e^{-\frac{x^2}{2}}.\end{aligned}$$

In the last step we could recognize the integral of the PDF of a $\mathcal{N}(x, 1)$ distributed random variable (as a function of y). From this we get

$$\begin{aligned}\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-\frac{x^2}{2} - \frac{(x-y)^2}{2}} dy dx &= \int_{-\infty}^{\infty} \sqrt{2\pi} e^{-\frac{x^2}{2}} dx \\ &= 2\pi \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx = 2\pi.\end{aligned}$$

In the last step we integrated the PDF of the standard normal. Hence, $c = \frac{1}{2\pi}$.

(b) We have basically computed f_X (without the constant c) in part (a) already.

$$\begin{aligned}f_X(x) &= \int_{-\infty}^{\infty} \frac{1}{2\pi} e^{-\frac{x^2}{2} - \frac{(x-y)^2}{2}} dy \\ &= \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{(x-y)^2}{2}} dy = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}.\end{aligned}$$

Now we compute f_Y :

$$f_Y(y) = \int_{-\infty}^{\infty} \frac{1}{2\pi} e^{-\frac{x^2}{2} - \frac{(x-y)^2}{2}} dx = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2} - \frac{(x-y)^2}{2}} dx.$$

We can complete the square in the exponent of the exponential:

$$-\frac{x^2}{2} - \frac{(x-y)^2}{2} = -(x^2 - xy - \frac{1}{2}y^2) = -(x - y/2)^2 - y^2/4,$$

and we can now compute the integral:

$$\begin{aligned}f_Y(y) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2} - \frac{(x-y)^2}{2}} dx \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-(x-y/2)^2 - y^2/4} dx \\ &= \frac{1}{\sqrt{4\pi}} e^{-y^2/4} \int_{-\infty}^{\infty} \frac{1}{\sqrt{\pi}} e^{-(x-y/2)^2} dx = \frac{1}{\sqrt{4\pi}} e^{-y^2/4}.\end{aligned}$$

In the last step we used the fact that $\frac{1}{\sqrt{\pi}} e^{-(x-y/2)^2}$ is the PDF of a $\mathcal{N}(y/2, 1)$ distributed random variable (as a function of x). It follows that $Y \sim \mathcal{N}(0, 2)$.

Thus $X \sim \mathcal{N}(0, 1)$ and $Y \sim \mathcal{N}(0, 2)$.

(c) X and Y are not independent since their joint density function is not the same as the product of the marginal densities.

Homework 11 – Solutions

8.6 (d) Since $X \sim \text{Geom}(p)$ and $Y \sim \text{Binom}(n, r)$, we know that

$$E[X] = \frac{1}{p}, \quad \text{Var}(X) = \frac{1-p}{p^2}, \quad E[Y] = nr, \quad \text{Var}(Y) = nr(1-r).$$

From these we can compute the second moments:

$$E[X^2] = \text{Var}(X) + (E[X])^2 = \frac{2-p}{p^2}, \quad E[Y^2] = \text{Var}(Y) + (E[Y])^2 = n(n-1)r^2 + nr.$$

To compute $E[(X+Y)^2]$ we first expand the square and use linearity:

$$E[(X+Y)^2] = E[X^2 + 2XY + Y^2] = E[X^2] + 2E[XY] + E[Y^2].$$

We already have $E[X^2]$ and $E[Y^2]$. For $E[XY]$ we use that these random variables are independent and hence

$$E[XY] = E[X]E[Y].$$

This gives

$$\begin{aligned} E[(X+Y)^2] &= E[X^2] + 2E[XY] + E[Y^2] \\ &= \frac{2-p}{p^2} + \frac{2nr}{p} + n(n-1)r^2 + nr. \end{aligned}$$

Another approach: Since the random variables are independent we have $\text{Var}(X+Y) = \text{Var}(X) + \text{Var}(Y)$. This gives

$$\begin{aligned} E[(X+Y)^2] &= \text{Var}(X+Y) + (E[X+Y])^2 = \text{Var}(X) + \text{Var}(Y) + (E[X] + E[Y])^2 \\ &= \frac{1-p}{p^2} + nr(1-r) + (1/p + nr)^2, \end{aligned}$$

which (after some algebra) leads to the same expression as the previous solution.

8.12 (a) We need to compute

$$M_Z(t) = E(e^{tZ}) = \int_{-\infty}^{\infty} e^{tz} f_Z(z) dz = \int_0^{\infty} e^{tz} \lambda^2 z e^{-\lambda z} dz = \lambda^2 \int_0^{\infty} z e^{-(\lambda-t)z} dz.$$

If $\lambda - t \leq 0$ then this integral is at least as large as $\lambda^2 \int_0^{\infty} z dz$ which is infinite. If $\lambda - t > 0$ then we can compute the integral using integration by parts to give

$$M_Z(t) = \begin{cases} \frac{\lambda^2}{(\lambda-t)^2}, & \text{if } t < \lambda \\ \infty, & \text{if } t \geq \lambda. \end{cases}$$

(b) We have already computed the MGF of an exponential distribution (see e.g. Example 5.6), so

$$M_X(t) = M_Y(t) = \begin{cases} \frac{\lambda}{\lambda-t}, & \text{if } t < \lambda \\ \infty, & \text{if } t \geq \lambda. \end{cases}$$

Since X and Y are independent, we have $M_{X+Y}(t) = M_X(t)M_Y(t)$. Comparing with part (a) we see that $X+Y$ has the same moment generating function as Z , which means that they must have the same distribution. (Since the moment generating function is finite in a neighborhood of 0.)

8.22 There are several ways to approach this problem. One possibility that gives the answer without doing complicated computations is as follows. For each $1 \leq j \leq 89$ let I_j be the indicator of the event that both j and $j + 1$ are chosen among the five numbers. Then $X = \sum_{j=1}^{89} I_j$, since if j and $j + 1$ are both chosen then they will be next to each other in the ordered sample. By linearity

$$E[X] = E\left[\sum_{j=1}^{89} I_j\right] = \sum_{j=1}^{89} E[I_j].$$

We can compute $E[I_j]$ directly by counting favorable outcomes:

$$E[I_j] = P(\text{both } j \text{ and } j + 1 \text{ are chosen}) = \frac{\binom{88}{3}}{\binom{90}{5}} = \frac{2}{801}.$$

Thus

$$E[X] = 89 \cdot \frac{2}{90 \times 89} = \frac{20}{90} = \frac{2}{9}.$$

Here is another possible solution: for any $1 \leq a < b \leq 5$ let $J_{a,b}$ be the indicator that the a th and b th picks are next to each other. (Note that the original 5 numbers are chosen with order!) Then $X = \sum_{1 \leq a < b \leq 5} J_{a,b}$ and because of the exchangeability of the 5 picks we have $E[J_{1,2}] = E[J_{a,b}]$ for any $1 \leq a < b \leq 5$. Hence by the linearity of expectation we have

$$E[X] = \sum_{1 \leq a < b \leq 5} E[J_{a,b}] = \binom{5}{2} E[J_{1,2}]$$

We have

$$E[J_{1,2}] = P(\text{the first two picks are next to each other}).$$

This probability can be computed by just considering the first two picks (and disregarding order) to get

$$E[J_{1,2}] = \frac{89}{\binom{90}{2}} = \frac{2}{90} = \frac{1}{45}$$

(since out of the equally likely $\binom{90}{2}$ pairs there are exactly 89 where we have two neighboring numbers). This gives

$$E[X] = \binom{5}{2} \frac{1}{45} = \frac{10}{45} = \frac{2}{9}.$$

8.23 (a) For $i = 1, \dots, 50$, let

$$Y_i = \begin{cases} 0, & \text{the } i\text{th pick is red;} \\ 1, & \text{the } i\text{th pick is green} \end{cases}$$

Then $Y_1, \dots, Y_{50}$ are exchangeable so

$$\begin{aligned} P(Y_{28} \neq Y_{29}) &= P(Y_{28} = 0, Y_{29} = 1) + P(Y_{29} = 0, Y_{28} = 1) \\ &= 2P(Y_1 = 0, Y_2 = 1) = 2 \left(\frac{20 \cdot 30}{50 \cdot 49} \right) = \frac{24}{49}. \end{aligned}$$

(b) Let I_j be the indicator that $Y_j \neq Y_{j+1}$ for $j = 1, \dots, 49$. Then $X = I_1 + \dots + I_{49}$ and by linearity

$$E[X] = \sum_{i=1}^{49} E[I_i] = \sum_{i=1}^{49} P(Y_i \neq Y_{i+1})$$

By the exchangeability of the Y_i random variables and part (a) we get

$$E[X] = \sum_{i=1}^{49} P(Y_i \neq Y_{i+1}) = 49P(Y_1 \neq Y_2) = 49 \left(\frac{24}{49} \right) = 24.$$

8.36 (a) Let I_k be the indicator of the event that the number k appears at least once among the four die rolls. Then $X = I_1 + \dots + I_6$ and we get

$$E[X] = E[I_1 + \dots + I_6] = E[I_1] + \dots + E[I_6] = 6E[I_1],$$

where the last step comes from exchangeability. We have

$$E[I_1] = P(\text{the number 1 shows up}) = 1 - P(\text{none of the rolls are equal to 1}) = 1 - \left(\frac{5}{6} \right)^4$$

which gives

$$E[X] = 6 \left(1 - \left(\frac{5}{6} \right)^4 \right).$$

(b) We can compute the variance either by computing $E[X^2]$ (and using part (a)) or using the formula for the variance of the sum. We will use the second approach.

We have $X = I_1 + \dots + I_6$, so

$$\text{Var}(X) = \text{Var}\left(\sum_{i=1}^6 I_i\right) = \sum_{i=1}^6 \sum_{j=1}^6 \text{Cov}(I_i, I_j).$$

Considering the $i = j$ and $i \neq j$ terms separately:

$$\text{Var}(X) = \sum_{i=1}^6 \text{Var}(I_i) + \sum_{i \neq j} \text{Cov}(I_i, I_j)$$

By exchangeability $\text{Var}(I_i) = \text{Var}(I_1)$ for all i , and $\text{Cov}(I_i, I_j) = \text{Cov}(I_1, I_2)$ for all $i \neq j$. Counting the respective terms gives

$$\text{Var}(X) = 6 \text{Var}(I_1) + 30 \text{Cov}(I_1, I_2).$$

The variance of a Bernoulli(p) random variable is $p(1 - p)$, so using part (a) we get

$$\text{Var}(I_1) = \left(1 - \left(\frac{5}{6} \right)^4 \right) \left(\frac{5}{6} \right)^4.$$

Recall that for two indicator random variables I_A, I_B (where A and B are the respective events) we have

$$\text{Cov}(I_A, I_B) = P(AB) - P(A)P(B).$$

To compute $\text{Cov}(I_1, I_2)$ the corresponding events are

$$A = \{\text{the number 1 shows up}\}, \quad B = \{\text{the number 2 shows up}\}.$$

We have seen in (a) that $P(A) = P(B) = \left(1 - \left(\frac{5}{6}\right)^4\right)$. To compute $P(AB)$ we consider the complement $P((AB)^c) = P(A^c \cup B^c)$ and use inclusion-exclusion:

$$P(A^c \cup B^c) = P(A^c) + P(B^c) - P(A^c B^c) = \left(\frac{5}{6}\right)^4 + \left(\frac{5}{6}\right)^4 - \left(\frac{2}{3}\right)^4$$

where we used the fact that $A^c B^c$ is the event that neither 1 or 2 shows up among the numbers. This gives

$$P(AB) = 1 - 2\left(\frac{5}{6}\right)^4 + \left(\frac{2}{3}\right)^4$$

and

$$\text{Cov}(I_1, I_2) = 1 - 2\left(\frac{5}{6}\right)^4 + \left(\frac{2}{3}\right)^4 - \left(1 - \left(\frac{5}{6}\right)^4\right)^2 = \left(\frac{2}{3}\right)^4 - \left(\frac{5}{6}\right)^8$$

Collecting everything gives

$$\text{Var}(X) = 6\left(1 - \left(\frac{5}{6}\right)^4\right)\left(\frac{5}{6}\right)^4 + 30\left(\left(\frac{2}{3}\right)^4 - \left(\frac{5}{6}\right)^8\right) \approx 0.447.$$

Alternate solution if you want to compute $\text{Var}(X)$ via $E[X^2]$:

$$\begin{aligned} E[X^2] &= E[(I_1 + \cdots + I_6)^2] = E\left[\sum_{k=1}^6 I_k^2 + 2 \sum_{j < k \leq 6} I_j I_k\right] \\ &= \sum_{k=1}^6 E[I_k^2] + 2 \sum_{j < k \leq 6} E[I_j I_k]. \end{aligned}$$

Since I_k is either 0 or 1, we have $I_k^2 = I_k$. Using exchangeability

$$\begin{aligned} E[X^2] &= \sum_{k=1}^6 E[I_k^2] + 2 \sum_{j < k \leq 6} E[I_j I_k] \\ &= \sum_{k=1}^6 E[I_k] + 2 \sum_{j < k \leq 6} E[I_j I_k] \\ &= 6E[I_1] + 30E[I_1 I_2]. \end{aligned}$$

From this point one can proceed as in the previous solution to get $E[I_1]$ (from part (a)) and $E[I_1 I_2]$ (as the probability $P(AB)$ above). This leads to the same answer:

$$E[X^2] = 6\left(1 - \left(\frac{5}{6}\right)^4\right) + 30\left(1 + \left(\frac{2}{3}\right)^4 - 2 \cdot \left(\frac{5}{6}\right)^4\right)$$

and

$$\begin{aligned} \text{Var}(X) &= E[X^2] - E[X]^2 \\ &= 6\left(1 - \left(\frac{5}{6}\right)^4\right) + 30\left(1 + \left(\frac{2}{3}\right)^4 - 2 \cdot \left(\frac{5}{6}\right)^4\right) - 36\left(1 - \left(\frac{5}{6}\right)^4\right)^2 \\ &\approx 0.447. \end{aligned}$$

8.40 (a) Let J_k be the indicator for the event that the number k is drawn in at least one of the 4 weeks. Then $X = J_1 + J_2 + \cdots + J_{90}$. Then by the linearity of expectation and exchangeability we get

$$E[X] = E\left[\sum_{k=1}^{90} J_k\right] = \sum_{k=1}^{90} E[J_k] = 90E[J_1].$$

We have

$$\begin{aligned}
E[J_1] &= P(1 \text{ is drawn at least one of the 4 weeks}) \\
&= 1 - P(1 \text{ is not drawn in any of the 4 weeks}) \\
&= 1 - \left(\frac{89 \cdot 88 \cdot 87 \cdot 86 \cdot 85}{90 \cdot 89 \cdot 88 \cdot 87 \cdot 86} \right)^4 = 1 - \left(\frac{85}{90} \right)^4.
\end{aligned}$$

From this

$$E[X] = 90E[J_1] = 90 \left(1 - \left(\frac{85}{90} \right)^4 \right) \approx 18.394.$$

8.42 We have

$$E[\bar{X}_n^4] = E \left[\left(\frac{X_1 + \cdots + X_n}{n} \right)^4 \right] = \frac{1}{n^4} E[(X_1 + \cdots + X_n)^4].$$

By expanding the fourth power of the sum and using linearity and exchangeability

$$\begin{aligned}
E[\bar{X}_n^4] &= \frac{1}{n^4} E \left[\sum_{k=1}^n X_k^4 + 24 \sum_{i < j < k < \ell} X_i X_j X_k X_\ell \right. \\
&\quad \left. + 12 \sum_{\substack{k < \ell \\ j \neq k, j \neq \ell}} X_j^2 X_k X_\ell + 6 \sum_{j < k} X_j^2 X_k^2 + 4 \sum_{j \neq k} X_j^3 X_k \right] \\
&= \frac{1}{n^4} \sum_{k=1}^n E[X_k^4] + 24 \sum_{i < j < k < \ell} E[X_i X_j X_k X_\ell] \\
&\quad + 12 \sum_{\substack{k < \ell \\ j \neq k, j \neq \ell}} E[X_j^2 X_k X_\ell] + 6 \sum_{j < k} E[X_j^2 X_k^2] + 4 \sum_{j \neq k} E[X_j^3 X_k] \\
&= \frac{1}{n^3} E[X_1^4] + 24 \binom{n}{4} E[X_1 X_2 X_3 X_4] \\
&\quad + 12 \cdot \binom{n}{3} E[X_1^2 X_2 X_3] + 6 \binom{n}{2} E[X_1^2 X_2^2] + 4n(n-1) E[X_1^3 X_2].
\end{aligned}$$

By independence

$$\begin{aligned}
E[X_1 X_2 X_3 X_4] &= E[X_1] E[X_2] E[X_3] E[X_4] = 0, & E[X_1^2 X_2 X_3] &= E[X_1^2] E[X_2] E[X_3] = 0, \\
E[X_1^3 X_2] &= E[X_1^3] E[X_2] = 0, & E[X_1^2 X_2^2] &= E[X_1^2] E[X_2^2] = E[X_1^2]^2 = a^2.
\end{aligned}$$

Hence

$$E[\bar{X}_n^4] = \frac{1}{n^3} E[X_1^4] + \frac{3n(n-1)}{n^4} E[X_1^2]^2 = \frac{c}{n^3} + \frac{3(n-1)a^2}{n^3}.$$

8.48 The joint probability mass function of the random variables (X, Y) can be represented by the

following table.

		Y		
		0	1	2
X	1	$\frac{9}{100}$	0	0
	2	$\frac{81}{100}$	$\frac{9}{100}$	0
	3	0	0	$\frac{1}{100}$

Hence, the marginal distribution are:

$$\begin{aligned} p_X(1) &= \frac{9}{100}, & p_X(2) &= \frac{90}{100}, & p_X(3) &= \frac{1}{100} \\ p_Y(0) &= \frac{90}{100}, & p_Y(1) &= \frac{9}{100}, & p_Y(2) &= \frac{1}{100}. \end{aligned}$$

From these we can compute the following expectations:

$$E[X] = \frac{48}{25}, \quad E[Y] = \frac{11}{100}, \quad E[XY] = \frac{6}{25},$$

and so

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y] = \frac{6}{25} - \frac{48}{25} \cdot \frac{11}{100} = \frac{18}{625}.$$

8.54 (a) We have

$$\begin{aligned} \text{Var}(X) &= E[X^2] - (E[X])^2 = 5 - 2^2 = 1 \\ \text{Var}(Y) &= E[Y^2] - (E[Y])^2 = 10 - 1^2 = 9 \\ \text{Cov}(X, Y) &= E[XY] - E[X]E[Y] = 1 - 2 \cdot 1 = -1. \end{aligned}$$

Then

$$\text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X) \text{Var}(Y)}} = \frac{-1}{\sqrt{1 \cdot 9}} = -\frac{1}{3}.$$

(b) We have

$$\text{Cov}(X, X + cY) = \text{Var}(X) + c \text{Cov}(X, Y) = 1 + c(-1) = 1 - c.$$

Thus for $c = 1$ the random variables X and $X + cY$ are uncorrelated.

Homework 10 – Solutions

- 7.2 The possible values for both X and Y are 0 and 1, hence $X + Y$ can take the values 0, 1 and 2. If $X + Y = 0$ then we must have $X = 0$ and $Y = 0$ and by independence we get

$$P(X + Y = 0) = P(X = 0, Y = 0) = P(X = 0)P(Y = 0) = (1 - p)(1 - r).$$

Similarly, if $X + Y = 2$ then we must have $X = 1$ and $Y = 1$:

$$P(X + Y = 2) = P(X = 1, Y = 1) = P(X = 1)P(Y = 1) = pr.$$

We can now compute $P(X + Y = 1)$ by considering the complement:

$$P(X + Y = 1) = 1 - P(X + Y = 0) - P(X + Y = 2) = 1 - (1 - p)(1 - r) - pr = p + r - 2pr.$$

We could have also used the fact that $X + Y = 1$ can only happen if $X = 0, Y = 1$ or $X = 1, Y = 0$:

$$P(X + Y = 1) = P(X = 0, Y = 1) + P(X = 1, Y = 0) = (1 - p)r + p(1 - r) = p + r - 2pr.$$

We have computed the probability mass function of $X + Y$ which identifies its distribution.

- 7.16 The probability mass function of X is

$$p_X(k) = \frac{\lambda^k}{k!} e^{-\lambda}, \quad k = 0, 1, 2, \dots$$

while the probability mass function of Y is $p_Y(0) = 1 - p$, $p_Y(1) = p$. Using the convolution formula we get

$$p_{X+Y}(n) = \sum_k p_X(k)p_Y(n-k).$$

The possible values of $X + Y$ are $0, 1, 2, \dots$, so we only need to deal with $n \geq 0$. We only have $p_Y(n - k) \neq 0$ if $n - k = 0$ or $n - k = 1$ so we get

$$p_{X+Y}(n) = p_X(n)p_Y(0) + p_X(n-1)p_Y(1).$$

If $n = 0$ then $p_X(n-1) = 0$, so

$$p_{X+Y}(0) = p_X(0)p_Y(0) = (1 - p)e^{-\lambda}.$$

For $n > 0$ we get

$$\begin{aligned} p_{X+Y}(n) &= p_X(n)p_Y(0) + p_X(n-1)p_Y(1) = (1 - p)\frac{\lambda^n}{n!}e^{-\lambda} + p\frac{\lambda^{n-1}}{(n-1)!}e^{-\lambda} \\ &= \lambda^{n-1}\frac{(\lambda(1-p) + np)}{n!}e^{-\lambda}. \end{aligned}$$

Thus the probability mass function of $X + Y$ is

$$p_{X+Y}(n) = \begin{cases} (1 - p)e^{-\lambda}, & \text{if } n = 0, \\ \lambda^{n-1}\frac{(\lambda(1-p) + np)}{n!}e^{-\lambda}, & \text{if } n \geq 1. \end{cases}$$

- 7.20 (b) Note that X takes values in $(0, 1)$, Y takes values in $(1, 2)$ so $X + Y$ will take values in $(1, 3)$. For a given $z \in (1, 3)$ the convolution formula gives

$$f_{X+Y}(z) = \int_{-\infty}^{\infty} f_X(x)f_Y(z-x)dx = \int_0^1 f_X(x)f_Y(z-x)dx,$$

where we used the fact that $f_X(x) = 0$ outside $(0, 1)$. For a given $1 < z < 3$ the function $f_Y(z-x)$ is nonzero if and only if $1 < z-x < 2$, which is equivalent to $z-2 < x < z-1$. Since we must have $0 < x < 1$ for $f_X(x)$ to be nonzero, this means that $f_X(x)f_Y(z-x)$ is nonzero only if $\max(0, z-2) < x < \min(1, z-1)$. Thus

$$\begin{aligned} f_{X+Y}(z) &= \int_0^1 f_X(x)f_Y(z-x)dx = \int_{\max(0, z-2)}^{\min(1, z-1)} 2x dx = x^2 \Big|_{\max(0, z-2)}^{\min(1, z-1)} \\ &= \min(1, z-1)^2 - \max(0, z-2)^2. \end{aligned}$$

We can evaluate the min and the max explicitly by separating the $1 < z \leq 2$ and $2 < z < 3$ cases. If $1 < z \leq 2$ then

$$\min(1, z-1) = z-1, \quad \max(0, z-2) = 0.$$

If $2 < z < 3$ then

$$\min(1, z-1) = 1, \quad \max(0, z-2) = z-2.$$

Hence

$$f_{X+Y}(z) = \begin{cases} (z-1)^2, & \text{if } 1 < z \leq 2, \\ 1 - (z-2)^2, & \text{if } 2 < z < 3, \\ 0, & \text{otherwise.} \end{cases}$$

- 7.24 Suppose that the variances of X, Y and Z are σ_X^2, σ_Y^2 and σ_Z^2 . Using Fact 7.9 about the linear combination of independent normal random variables we have that

$$X + 2Y - 3Z \sim \mathcal{N}(0, \sigma_X^2 + 4\sigma_Y^2 + 9\sigma_Z^2),$$

$$\text{and } \frac{X+2Y-3Z}{\sqrt{\sigma_X^2+4\sigma_Y^2+9\sigma_Z^2}} \sim \mathcal{N}(0, 1).$$

This gives

$$P(X + 2Y - 3Z > 0) = P\left(\frac{X + 2Y - 3Z}{\sqrt{\sigma_X^2 + 4\sigma_Y^2 + 9\sigma_Z^2}} > 0\right) = 1 - \Phi(0) = \frac{1}{2}.$$

Another way to see the answer is the following: if X, Y, Z are mean zero then they have the same distribution as $-X, -Y, -Z$, respectively. Then $X + 2Y - 3Z$ has the same distribution as $-X - 2Y + 3Z$ which means that

$$P(X + 2Y - 3Z > 0) = P(-X - 2Y + 3Z > 0) = P(X + 2Y - 3Z < 0).$$

Since $X + 2Y - 3Z$ is continuous, we have $P(X + 2Y - 3Z = 0) = 0$ and hence

$$P(X + 2Y - 3Z > 0) = P(X + 2Y - 3Z < 0) \rightsquigarrow P(X + 2Y - 3Z > 0) = 1/2.$$

7.28 Because X_1, X_2, X_3 are jointly continuous, the probability that any two of them are equal is 0. This means that $P(X_1, X_2, X_3 \text{ are all different}) = 1$. By the exchangeability of X_1, X_2, X_3 we have

$$\begin{aligned} P(X_1 < X_2 < X_3) &= P(X_2 < X_1 < X_3) = P(X_1 < X_3 < X_2) \\ &= P(X_3 < X_2 < X_1) = P(X_2 < X_3 < X_1) = P(X_3 < X_1 < X_2), \end{aligned}$$

where we listed all six possible orderings of X_1, X_2, X_3 . By additivity the sum of the six probabilities is $P(X_1, X_2, X_3 \text{ are all different})$. From this we get that $P(X_1 < X_2 < X_3) = \frac{1}{6}$.

7.30 We can label the cards of the deck with the integers from 1 to 52. Because we are sampling without replacement, the five random variables corresponding to the five picks are exchangeable.

(a) By exchangeability we can change the 2nd and 4th cards to 1st and 2nd in the event without changing the probability:

$$P(\text{2nd card is A, 4th card is K}) = P(\text{1st card is A, 2nd card is K}) = \frac{4 \cdot 4}{52 \cdot 51} = \frac{4}{663},$$

where the final probability comes from counting the favorable outcomes for the first two picks.

(b) Again, by exchangeability and counting the favorable outcomes within the first two picks:

$$P(\text{1st card is } \spadesuit, \text{ 5th card is } \spadesuit) = P(\text{1st card is } \spadesuit, \text{ 2nd card is } \spadesuit) = \frac{\binom{13}{2}}{\binom{52}{2}} = \frac{1}{17}.$$

(c) We use exchangeability again.

$$\begin{aligned} P(\text{2nd card is K} | \text{last two cards are aces}) &= \frac{P(\text{2nd card is K, last two cards are aces})}{P(\text{last two cards are aces})} \\ &= \frac{P(\text{3rd card is K, first two cards are aces})}{P(\text{first two cards are aces})} \\ &= P(\text{3rd card is K} | \text{first two cards are aces}) \\ &= \frac{4}{50} = \frac{2}{25}. \end{aligned}$$

The final probability comes either from counting favorable outcomes for the first three picks, or by noting that if we choose two aces for the first two picks then we always have 50 cards left with 4 of them being kings.